#FABCONSQLCON2026

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Microsoft Fabric COMMUNITY CONFERENCE | Microsoft SQL COMMUNITY CONFERENCE

ATLANTA MARCH 16 - 20, 2026

https://aka.ms/FabricCapacitiesWorkshop

Microsoft Fabric Capacities

Essentials for Smarter Operations

Sanjay Raut, Pat Mahoney, Matt Stenzel

Principal PM's - Azure Data CAT

Meet the Speakers and Proctors

- Speakers
  - Matt Stenzel
  - Sanjay Raut
  - Pat Mahoney

- Proctors
  - Tim Bindas
  - Pratik Patel

Full-Day Workshop Agenda Timeline

9:00 AM

Session 1 - Capacity Concepts (90 min)

CU model, bursting, throttling, hands-on lab [1]

10:30 AM

Session 2 - Recovery & Advanced Config (90 min)

Throttling recovery, advanced settings, licensing [1]

1:30 PM

Session 3 - Monitoring & Events (120 min)

Metrics App, Real-Time Hub, chargeback demos [1]

3:30 PM

Session 4 - Planning & Optimization (30 min)

Best practices, scale decisions, workload isolation [1]

4:00 PM

Session 5 - Q&A with Leadership (30 min)

Open forum with Capacity Engineering Leadership [1]

- Structured learning progression: From foundational concepts to advanced topics
- Interactive engagement with hands-on labs, live demonstrations,
- Bonus - dedicated Q&A sessions towards end of the day to ask further clarifications around real capacity scenarios
- Lunch Break:
  - 12:00 Noon to 1:30 PM

Session 2 – Recovery & Advance Config

Throttling recovery, advanced settings

What we covered so far

- Basic concept of capacity unit and its evolution from v-core to CU model
- Basic concept of bursting and smoothing
- Basic Concept of throttling (10min, 60 min, 24 Hr)

How to solve throttling

Immediate

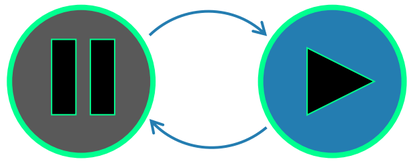
- Pause Resume
- Resize
- Capacity configurations
  - Overage billing
  - Surge Protection
  - Mission Critical
  - Spark Auto-scale

Long term

- Optimize
- Scale up
- Scale out
- Isolate

Pause and Resume

Pausing and Resuming Capacities



Why pause capacities?

1) It helps manage compute costs. (PAYG)
2) It clears any debt that has accumulated. Use it to quickly resolve throttling. (PAYG / RI)

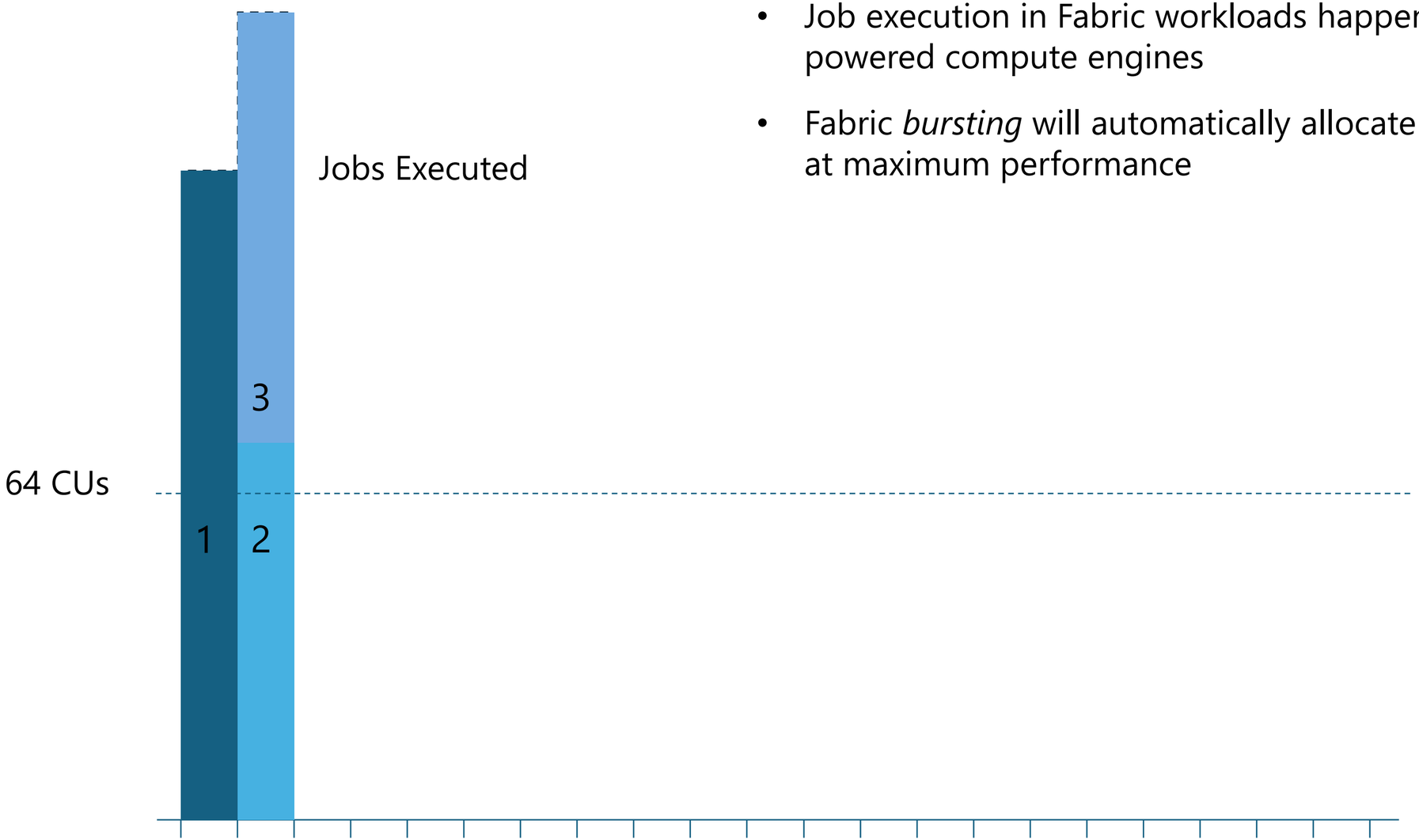
What does it do?

Workloads stop execution within 10 minutes of Pause action

New requests are not allowed to Start

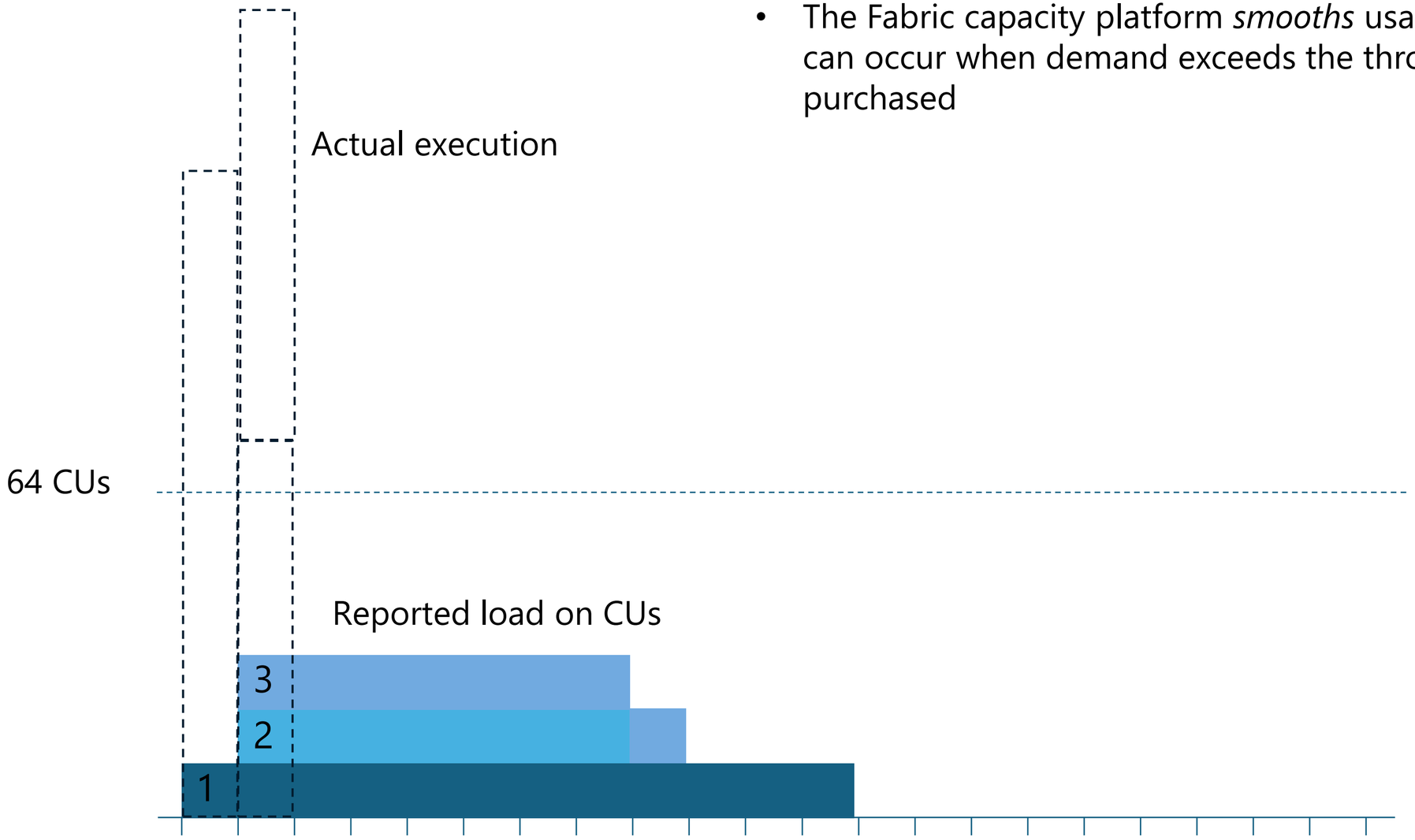
Smoothed usage will be reconciled

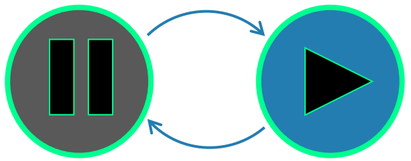
Note: OneLake storage costs continue to be billed while a capacity is paused



- Job execution in Fabric workloads happens on-demand via capacity powered compute engines
- Fabric *bursting* will automatically allocate resources as needed to execute at maximum performance

- The Fabric capacity platform *smooths* usage out to reduce throttling which can occur when demand exceeds the throughput of the capacity that was purchased

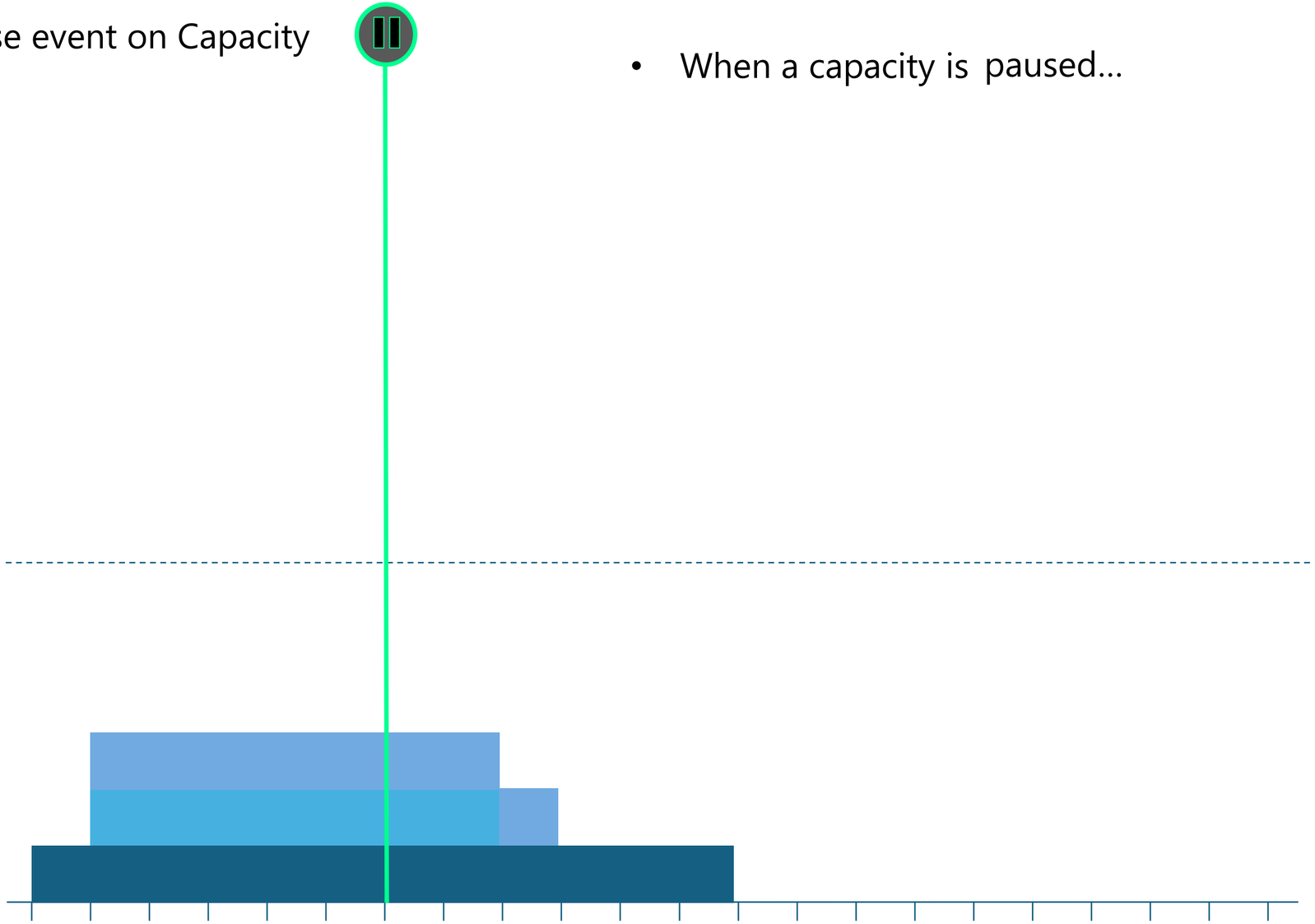


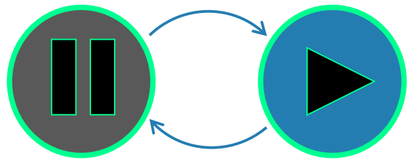


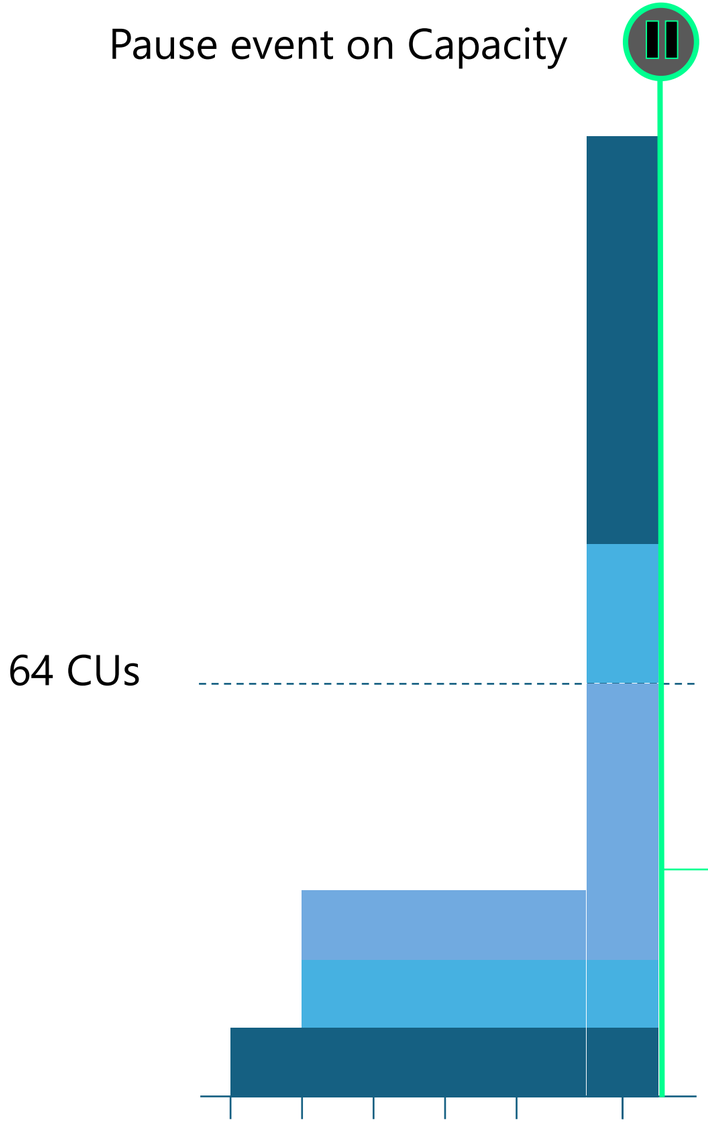
Pause event on Capacity

- When a capacity is paused…

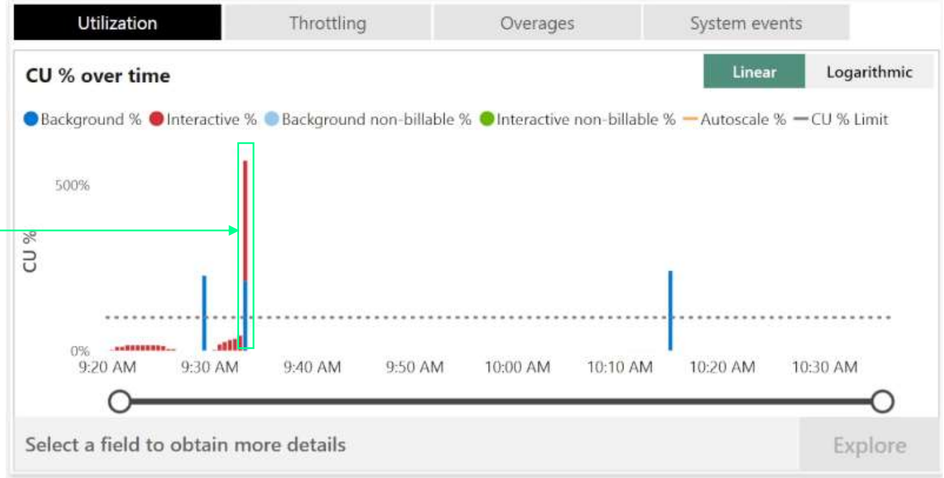
64 CUs







- When a capacity is paused…
- Usage that was smoothed into the future will be “reconciled” and charged against the capacity at the timestamp the capacity was paused
- Reconciled usage will show up as a spike in capacity metrics



Pausing a Fabric Capacity

It might actually cost you more ..

Don’t blindly pause

- Especially if you’re hoping it will reduce costs
- Be mindful of ‘open balance’
- How long would you need to pause for it to be beneficial?

Throttling

- In a throttled state, this can add up
- What is the price for business continuity?

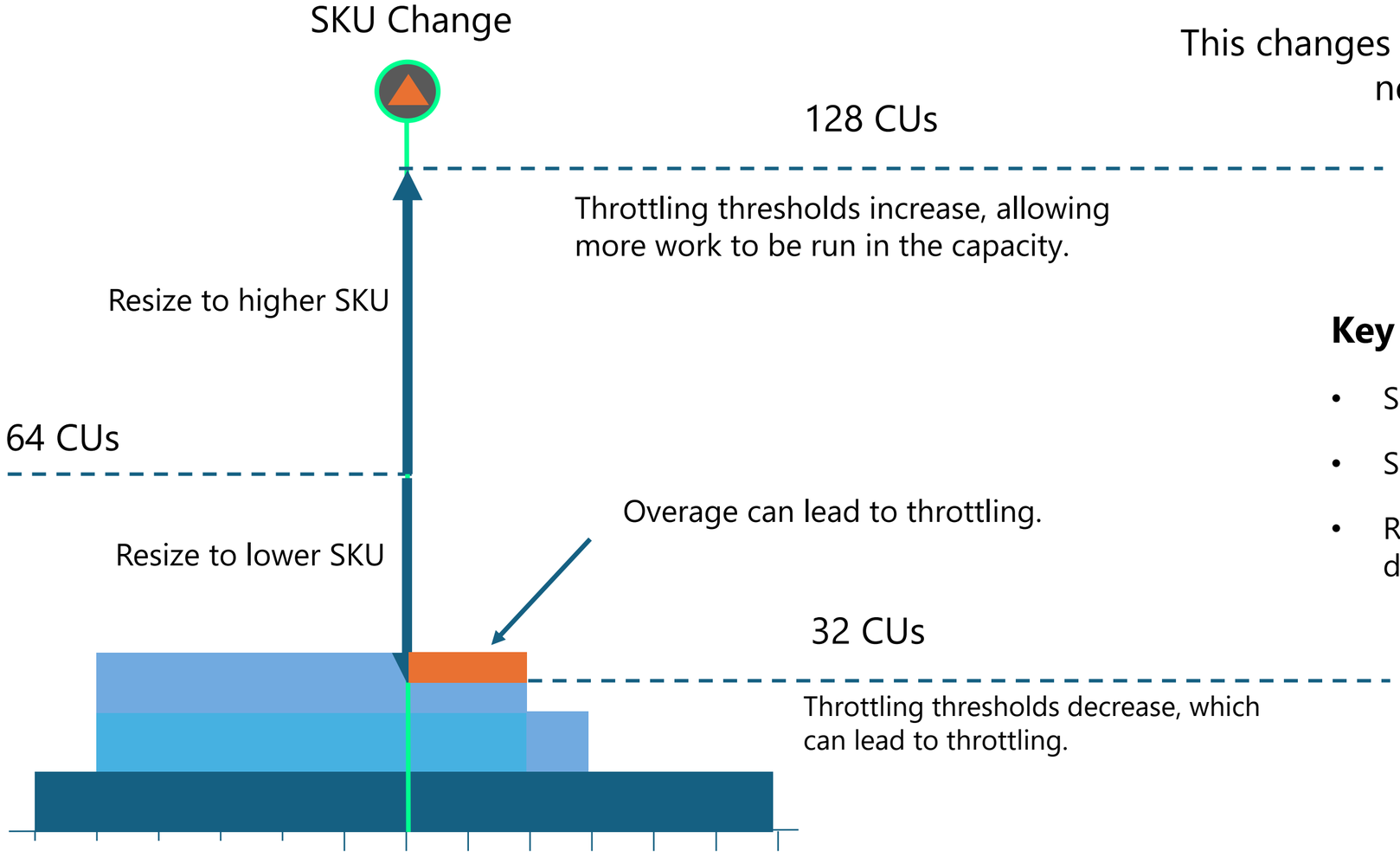
https://www.linkedin.com/pulse/fabric-billing-part-4-implications-pause-restart-matthew-farrow-fznse/

	Average Percentage Utilisation Next 24 Hours	Min Pause Hours for Saving	Cost of Pausing European F64
10,000%	3.47%	0.83	$10.13
20,000%	6.94%	1.67	$20.27
30,000%	10.42%	2.50	$30.40
40,000%	13.89%	3.33	$40.53
50,000%	17.36%	4.17	$50.67
60,000%	20.83%	5.00	$60.80
70,000%	24.31%	5.83	$70.93
80,000%	27.78%	6.67	$81.07
90,000%	31.25%	7.50	$91.20
100,000%	34.72%	8.33	$101.33
110,000%	38.19%	9.17	$111.47
120,000%	41.67%	10.00	$121.60
130,000%	45.14%	10.83	$131.73
140,000%	48.61%	11.67	$141.87
150,000%	52.08%	12.50	$152.00
160,000%	55.56%	13.33	$162.13
170,000%	59.03%	14.17	$172.27
180,000%	62.50%	15.00	$182.40
190,000%	65.97%	15.83	$192.53
200,000%	69.44%	16.67	$202.67
210,000%	72.92%	17.50	$212.80
220,000%	76.39%	18.33	$222.93
230,000%	79.86%	19.17	$233.07
240,000%	83.33%	20.00	$243.20
250,000%	86.81%	20.83	$253.33
260,000%	90.28%	21.67	$263.47
270,000%	93.75%	22.50	$273.60
280,000%	97.22%	23.33	$283.73
288,000%	100.00%	24.00	$291.84

Resize

Resizing Capacities

When a capacity is **resized**…



The allowed CUs per timepoint increase or decrease.

This changes the throttling allowed limits based on the new SKU's CUs and the throttling windows.

Key Insights

- Sizing up will incur the cost of the new SKU
- Sizing down could lead to more throttling
- Review your Throttling Thresholds before sizing down your SKU.

Scaling Capacities – know the border cases

Just be careful with this 🙃

- Scaling up a capacity that's smaller than F64 to a larger capacity happens almost immediately.
- **The capacity license updates usually take up to a day;** however, some updates could take longer.
- During this time, Fabric Free users might see a request to upgrade to a Power BI Pro license when they try to access a Power BI report.

https://learn.microsoft.com/en-us/fabric/enterprise/scale-capacity#considerations-and-limitations

Capacity configurations

- Overage Billing
- Surge Protection & Mission Critical
- Spark Auto-scale

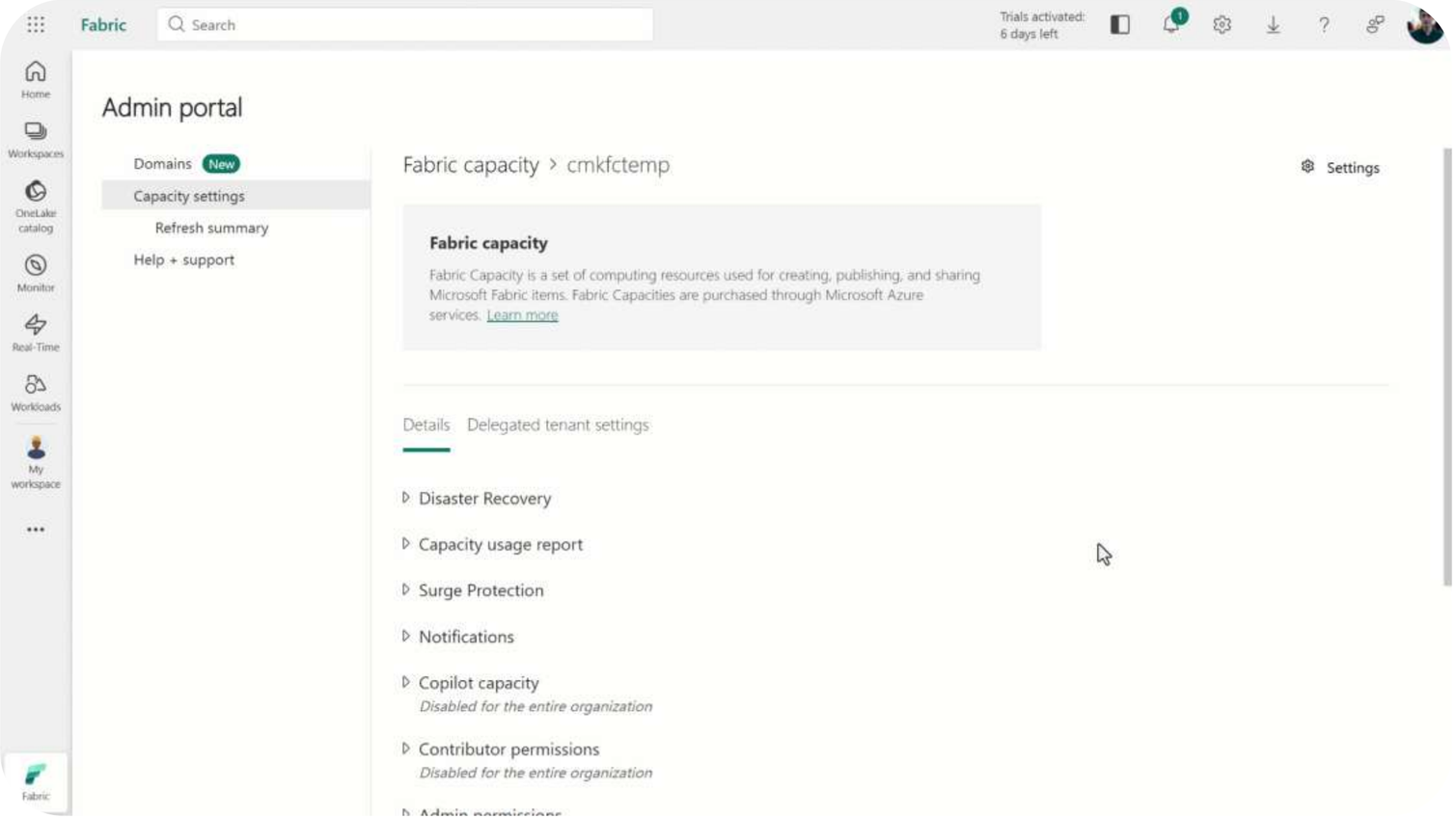
Capacity Overage Billing

Automatically prevent throttling during periods of peak demand

Pay only for extra usage when capacity limits are exceeded

Set spending limits directly in the Admin Portal

Right-size your SKU—no need to over-provision for spikes



Surge Protection v1 – Background usage limits

GA

Enabling Capacity Admins to get ahead of throttling. (blog || docs)

Simple experience that limits overuse by background jobs

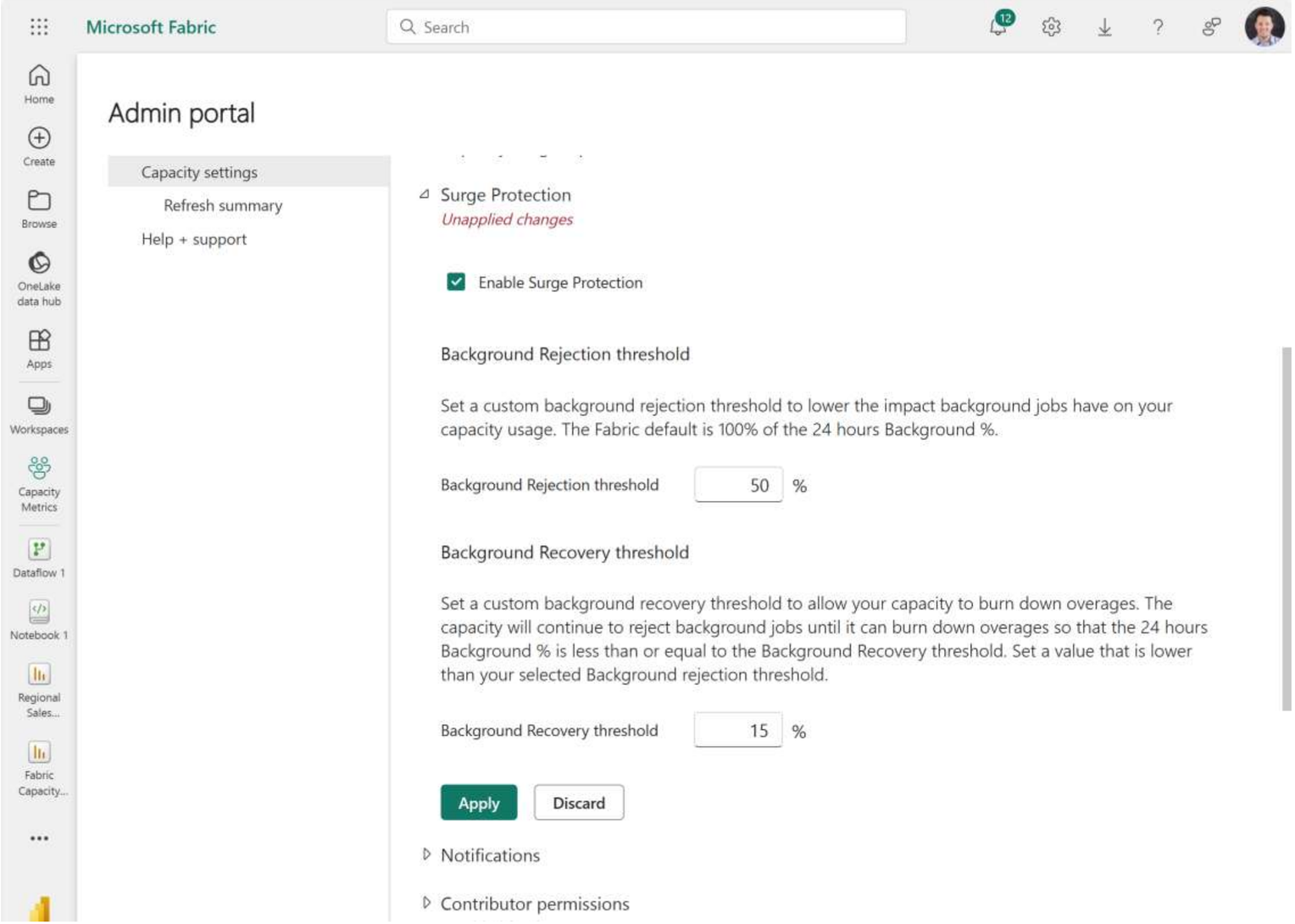
- Throttles background jobs before a full 24-hours of CUs is consumed.
- Helps to **protect interactive** usage like Power BI

Recovery limit

- Keep throttling until the capacity is 'healthy' as defined by the customer
- Helps prevent a capacity from immediately being throttled again

Impact

- Throttling background jobs will help 40-60% of capacities experiencing Interactive Rejections.



Surge Protection V2 – Workspace limits

Preview

Enabling Capacity Admins to get ahead of throttling.

Simple experience that limits overuse by workspaces

- **Blocks a workspaces** for 24-hours if it exceeds a limit set by the Capacity Admin
- **Mission critical workspaces** are excluded from surge protection limits

Self-healing

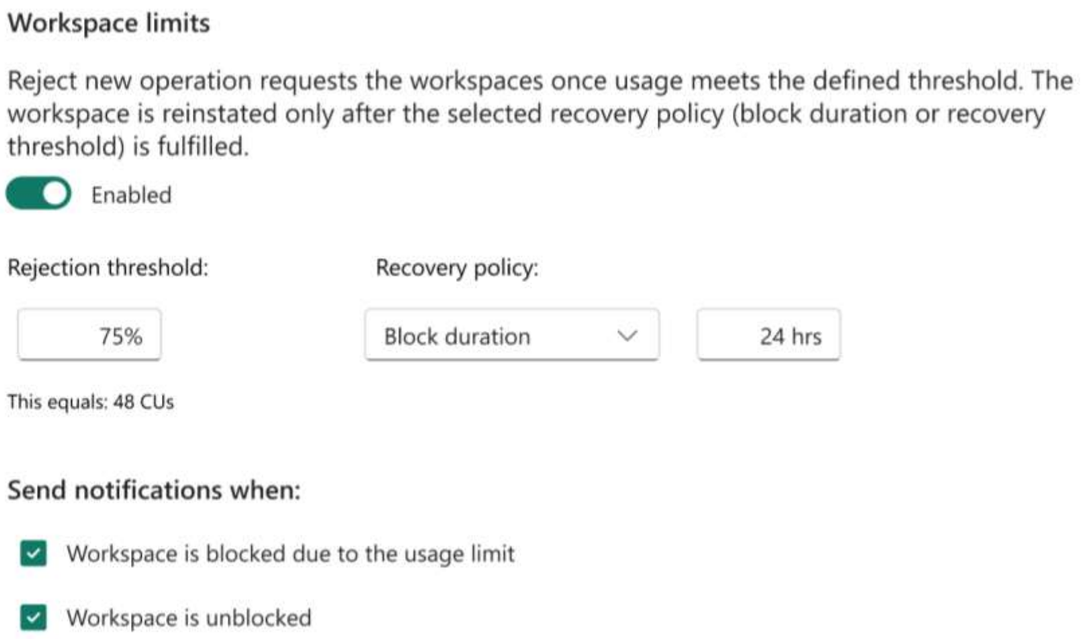
- Automatically restores workspace after 24 hours.

Capacity Admins are in control

- Can add / remove blocked workspaces
- APIs are provided

Impact

- Limit blocking to mis-behaving workspaces allowing the rest of the org to continue working without interruption.



Autoscale billing for Apache Spark

GA

Enable everyone in your org to use Spark and manage its cost (blog || docs)

Serverless style billing for Spark jobs

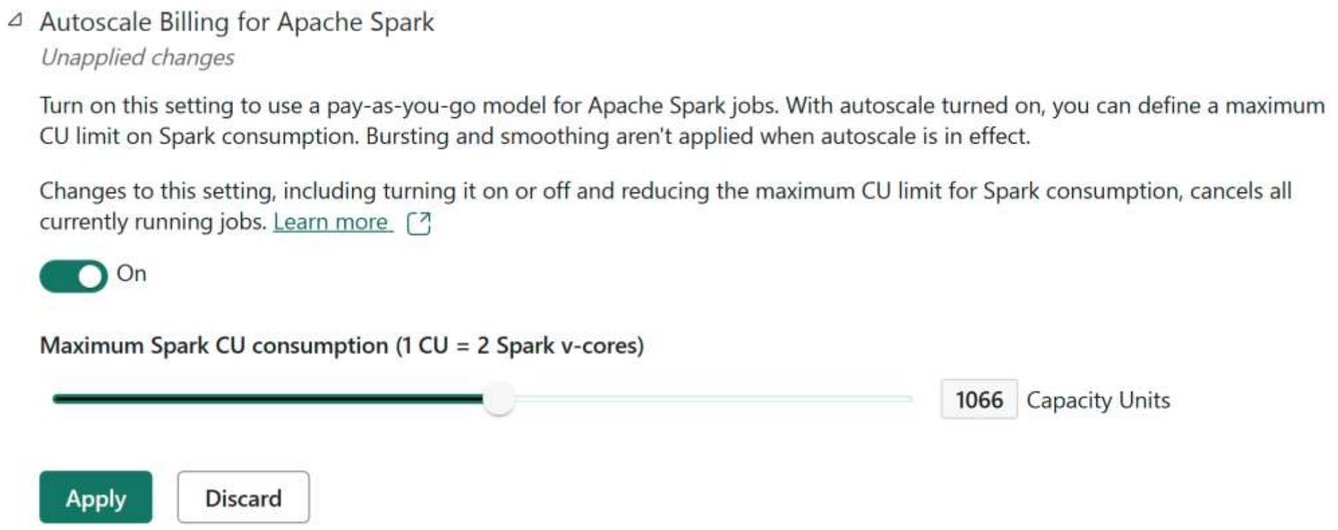
- Capacity admins can opt-in
- Set a max limit on CU used by Spark
- Only pay for what you use
- Spark manages the limit ensuring pools don't over consume
- Observability in a new metrics app page

Spark jobs are billed separately

- Jobs are billed when they execute
- Cost is at **Pay-as-you-go rate**
- Must also have an active capacity
- If Spark calls other workloads, like OneLake, those costs are billed to the capacity.

Impact

- Isolate your spiky Spark jobs from the rest of capacity compute
- Helps save on costs and reduce throttling



But .. Is it right for you?

- Part 1
- Part 2
- Part 3

Autoscale billing for Apache Spark

GA

Monitor consumption in Metrics app

New Autoscale Compute for Spark page

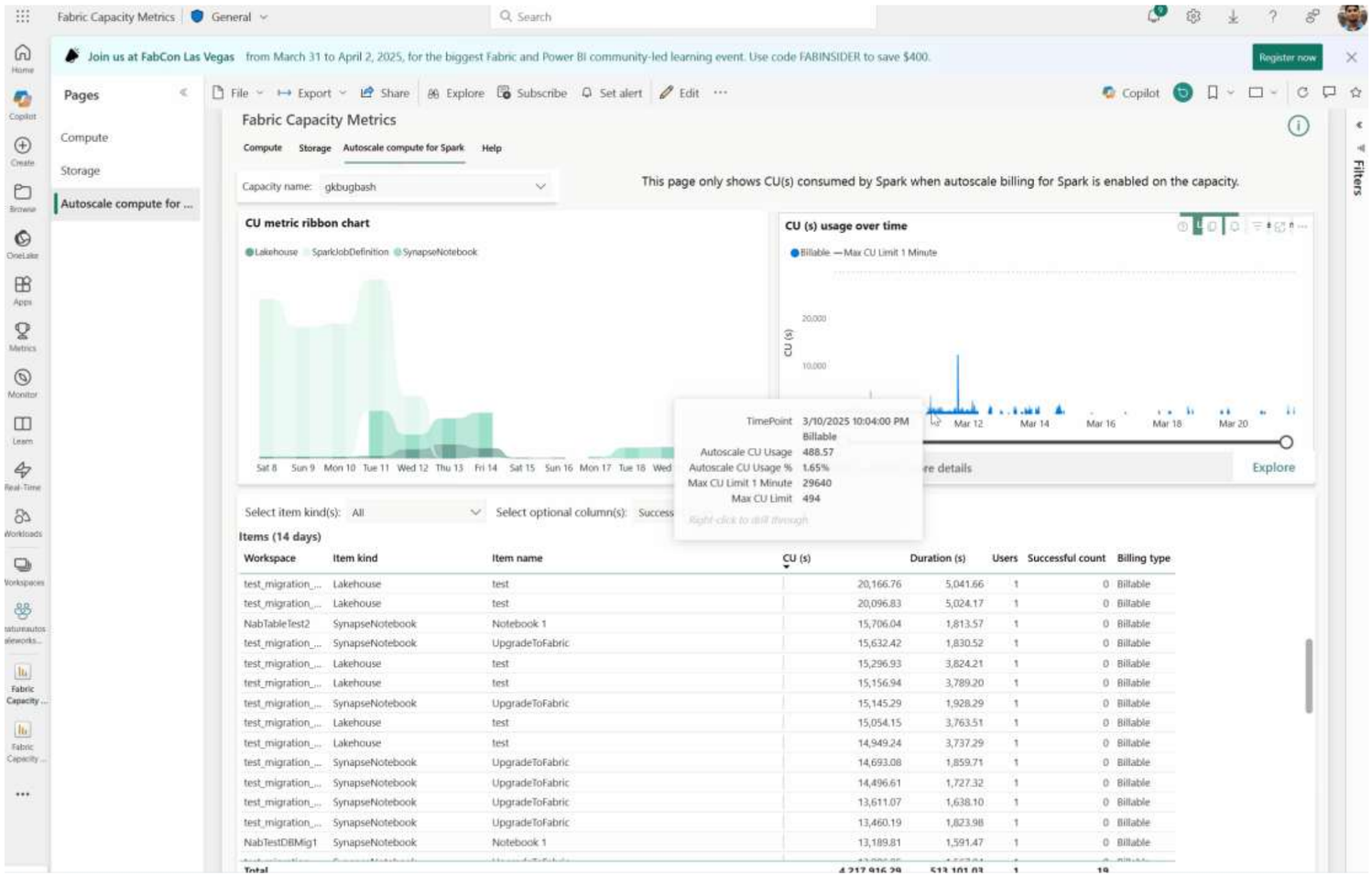
- Shows Spark CU consumed through Autoscale
- Easy to track to the configured autoscale limit.

Familiar experience

- Same experiences as for the capacity compute page.
- Provides drill down experience to see operation details

Impact

- Clearly understand the compute specific to Spark that will be reflected on your bill



Job-Level Bursting for Spark

GA

Disable Bursting for Spark Compute on a given capacity (blog || docs)

Capacity Level Switch for Bursting

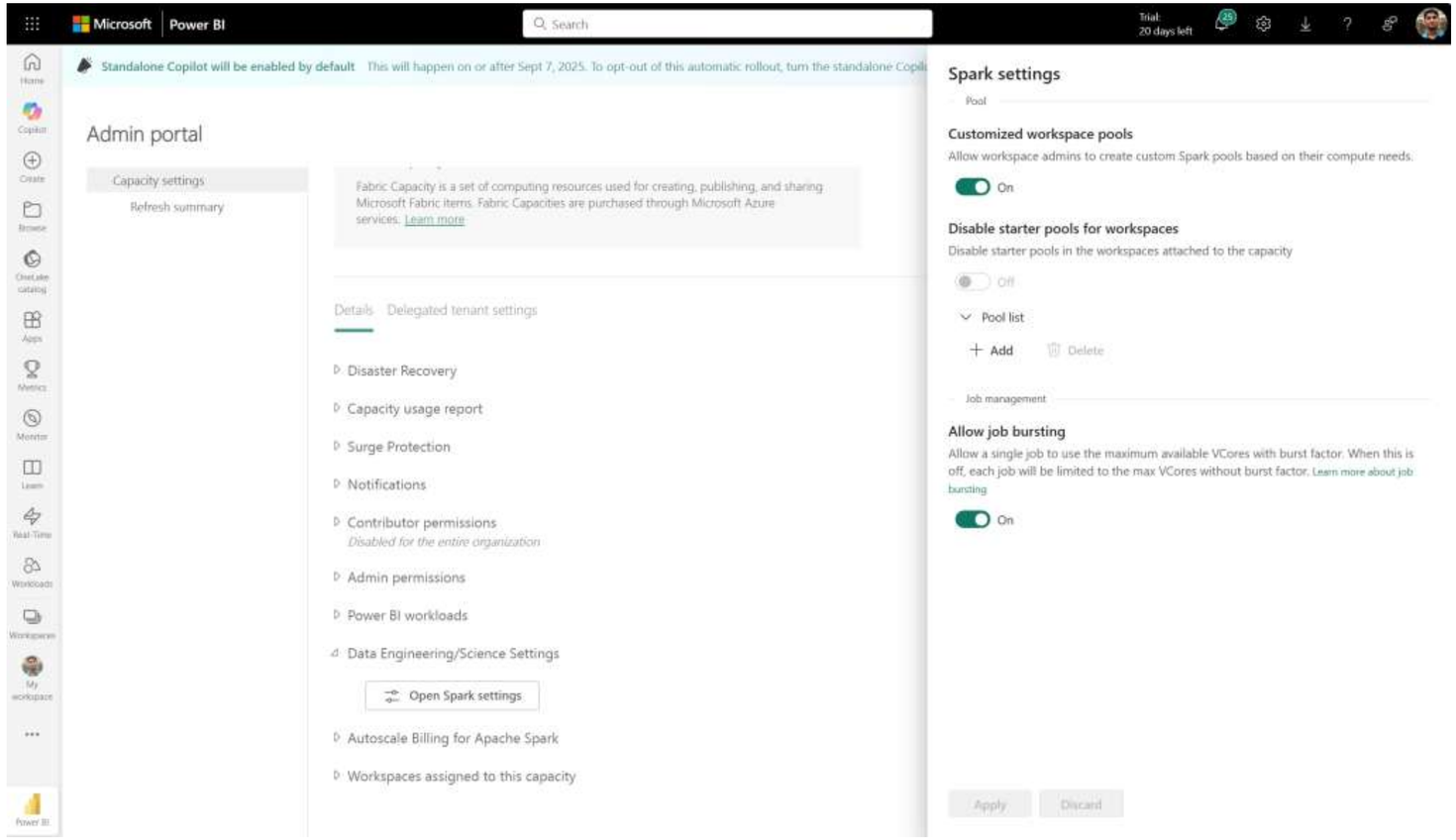
- Enabled (Default) – Up to 3x CUs allowed
- Disabled – No bursting, predictable concurrency

AutoScale Billing & Job-Level Bursting

- Cannot be combined!
- AutoScale Billing does not follow bursting

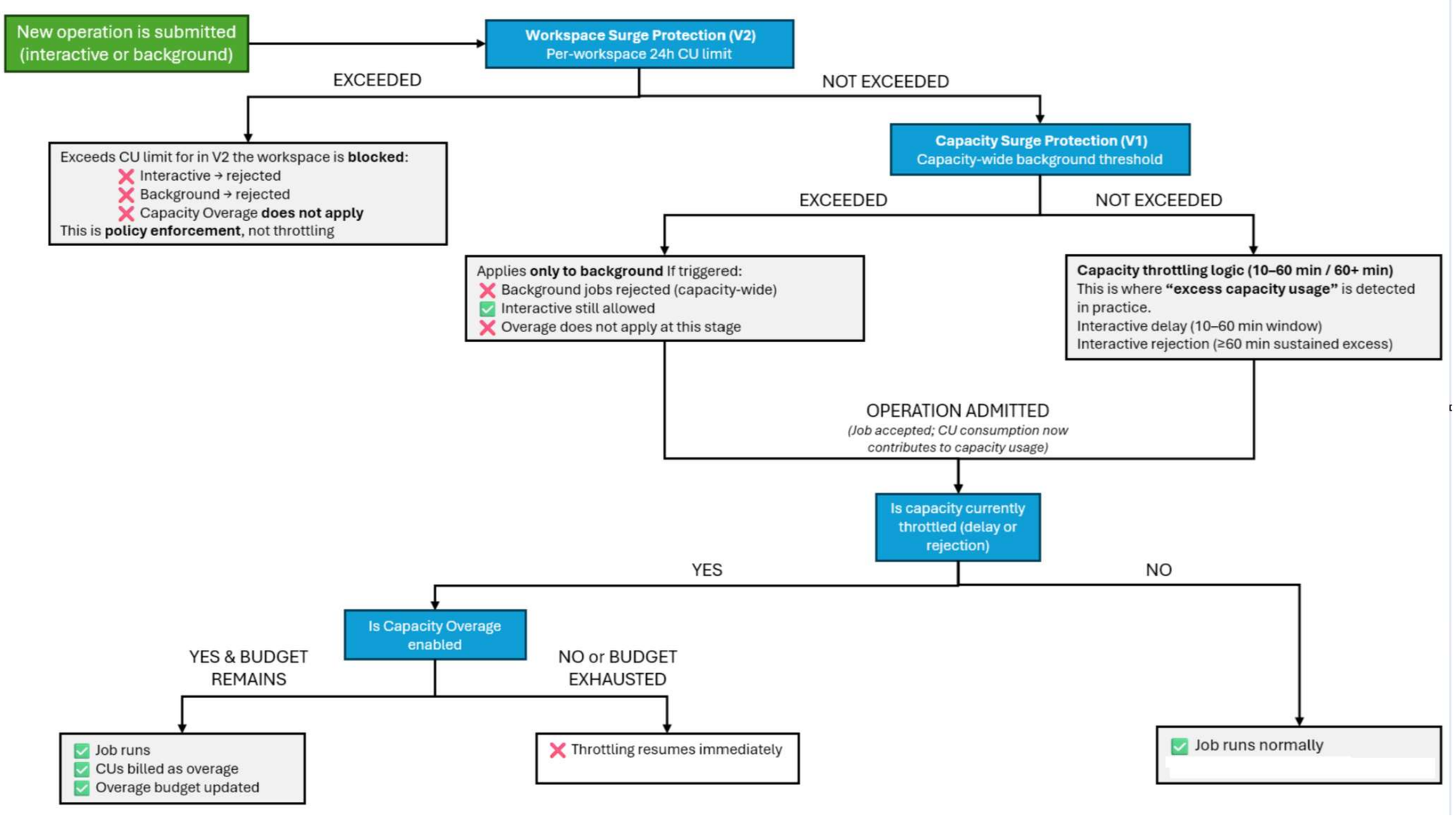
Impact

- Provide more predictable usage from Spark on your capacity, reducing spikes and helping to control concurrency



Putting All Configs together

The Process


New operation is submitted (interactive or background)
Workspace Surge Protection (V2)
Per-workspace 24h CU limit
EXCEEDED
NOT EXCEEDED
Exceeds CU limit for in V2 the workspace is blocked:
Interactive → rejected
Background → rejected
Capacity Overage does not apply
This is policy enforcement, not throttling
Capacity Surge Protection (V1)
Capacity-wide background threshold
EXCEEDED
NOT EXCEEDED
Applies only to background If triggered:
Background jobs rejected (capacity-wide)
Interactive still allowed
Overage does not apply at this stage
Capacity throttling logic (10–60 min / 60+ min)
This is where "excess capacity usage" is detected in practice.
Interactive delay (10–60 min window)
Interactive rejection (≥60 min sustained excess)
OPERATION ADMITTED
(Job accepted; CU consumption now contributes to capacity usage)
Is capacity currently throttled (delay or rejection)
YES
NO
Is Capacity Overage enabled
YES & BUDGET REMAINS
NO or BUDGET EXHAUSTED
Job runs
CUs billed as overage
Overage budget updated
Throttling resumes immediately
Job runs normally

Long Term

- Optimize
- Scale Up / Scale Out
- Isolate

Provisioning and Deploying Capacities

Tenant -> Region -> Capacity -> Workspace -> items

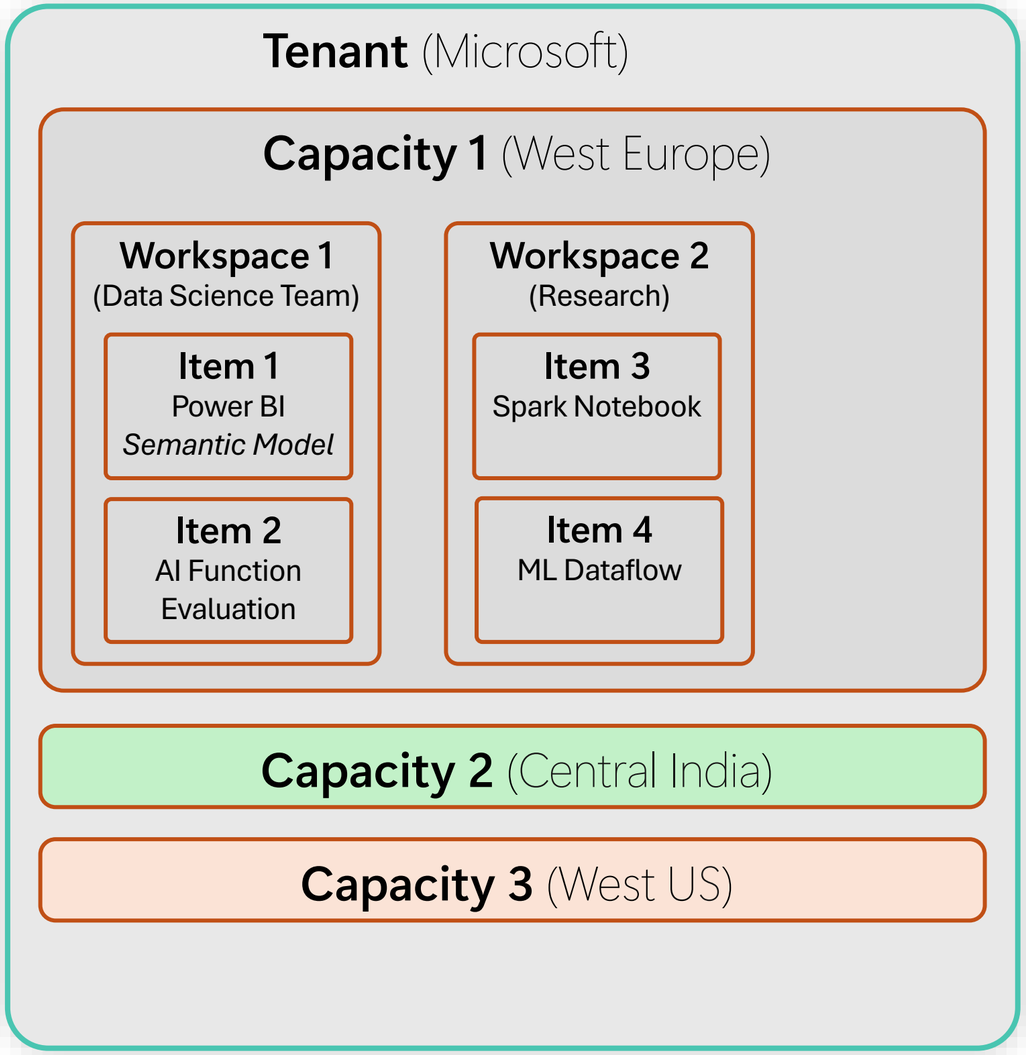
- Each capacity **resides in a specific region of the buyers' choice** where both the data & compute reside

Data residency defined by capacity

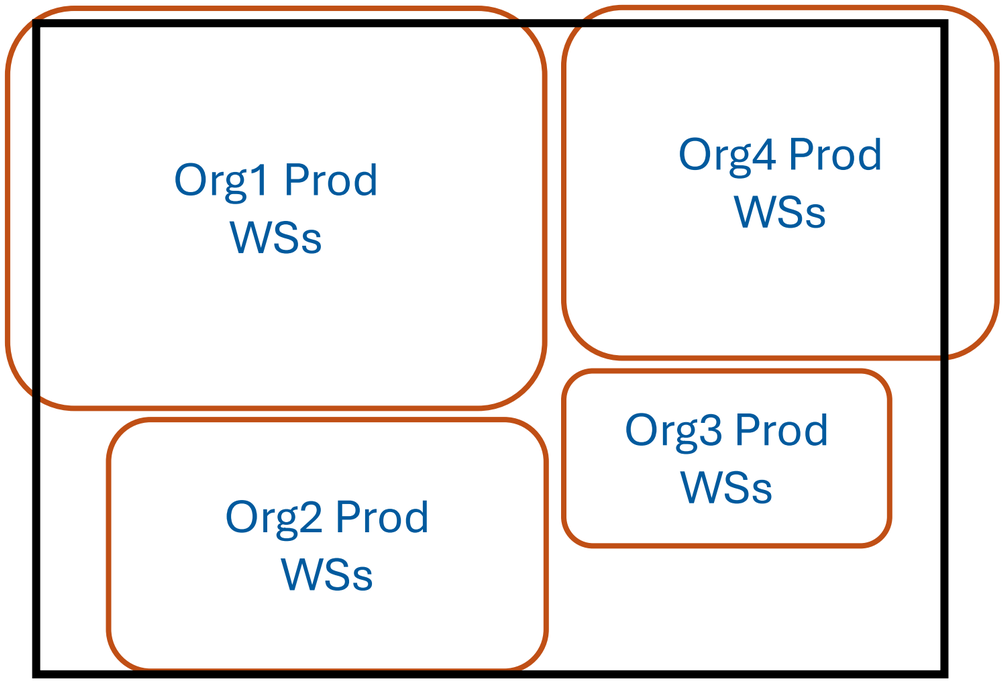
- **Workspaces are assigned to a capacity** that provides the compute and storage for all the workspace artifacts

Billing defined by Capacities

- Multiple capacities can be purchased, deployed and managed by **different owners** residing in a single tenant allowing each business unit to pay for their own consumption



Optimize



WSs = Workspaces

Capacity

Approach

- Work with content creators to follow best practices and reduce CU consumption

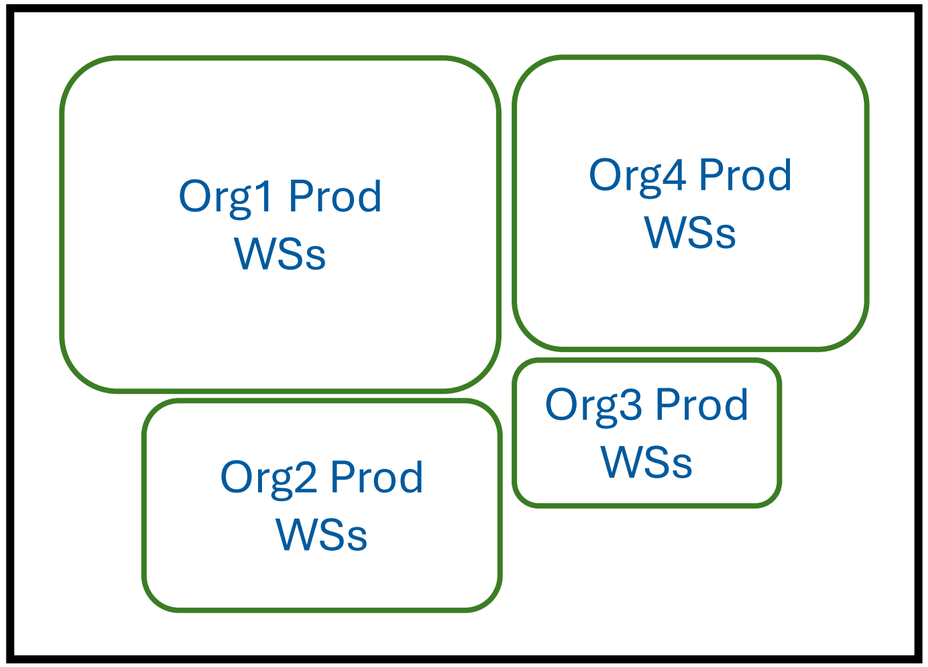
Pros

- Avoids increased cost
- Learning carries over to future content

Cons

- Can be difficult/time consuming

Optimize



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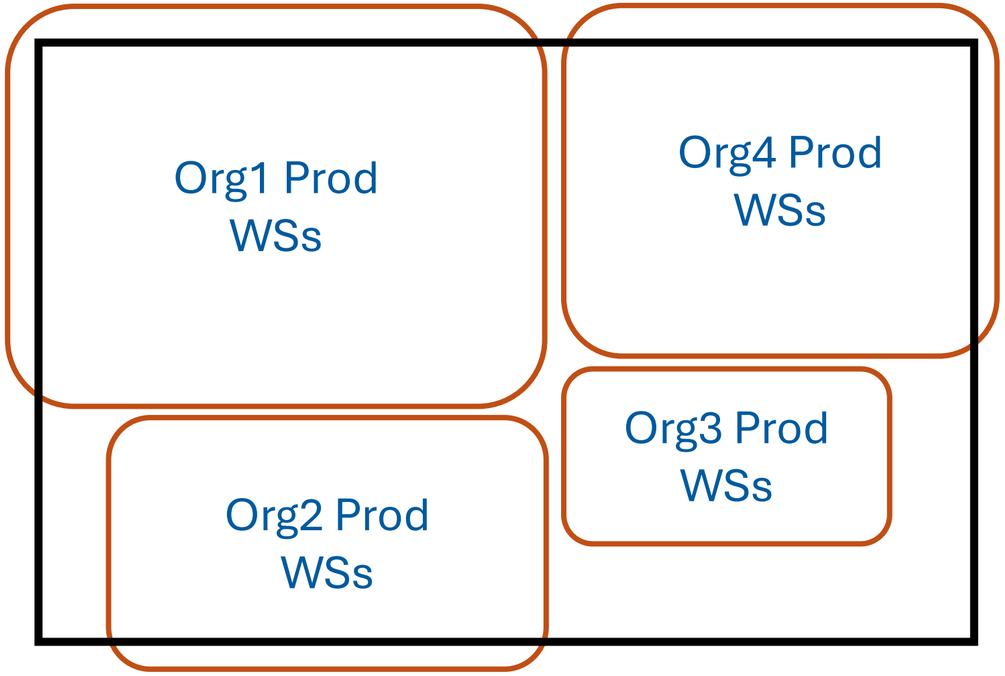


WSs = Workspaces Capacity

Options to add compute

- Move to a bigger F SKU
- Schedule Scale using Azure automation, Fabric CLI or notebook

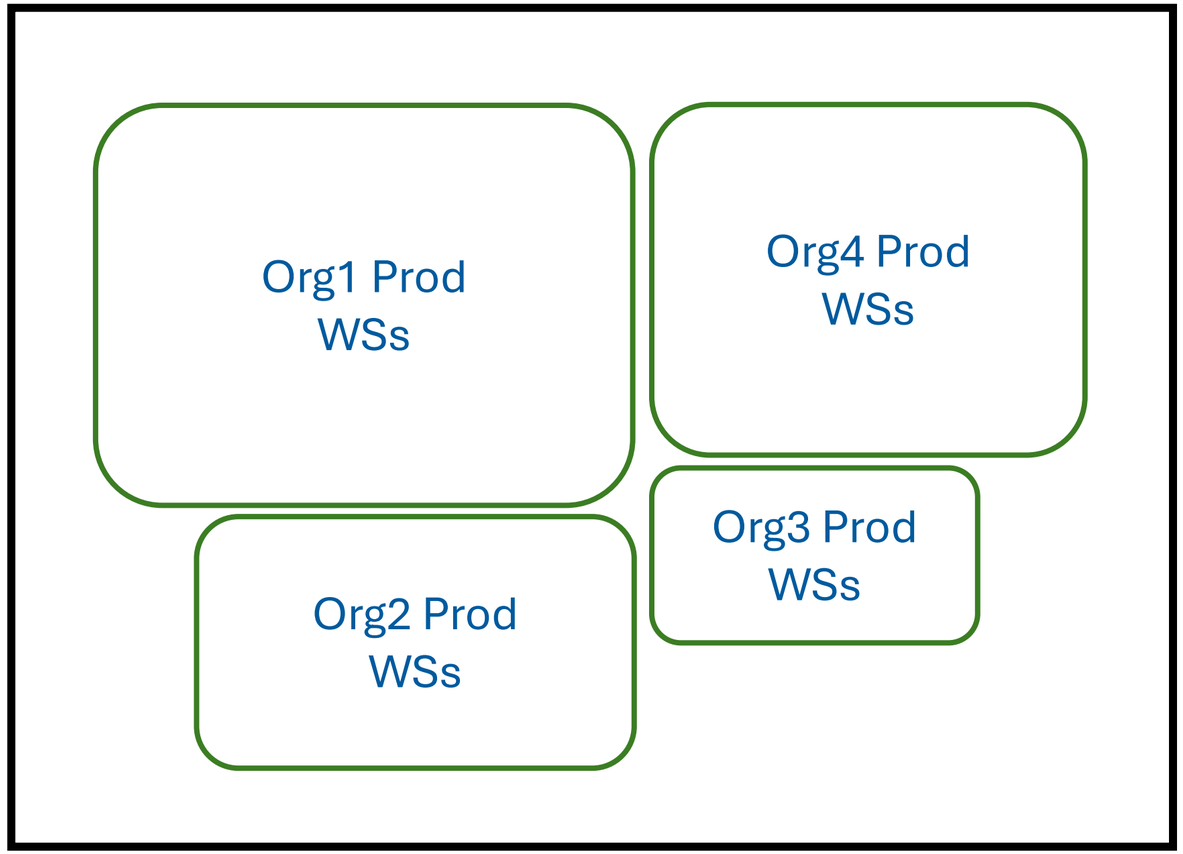
Pros

- Add CUs for all items
- Easy

Cons

- Cost
- Noisy neighbors (items with unintentionally high CU burn) can still be a problem

Scale Up



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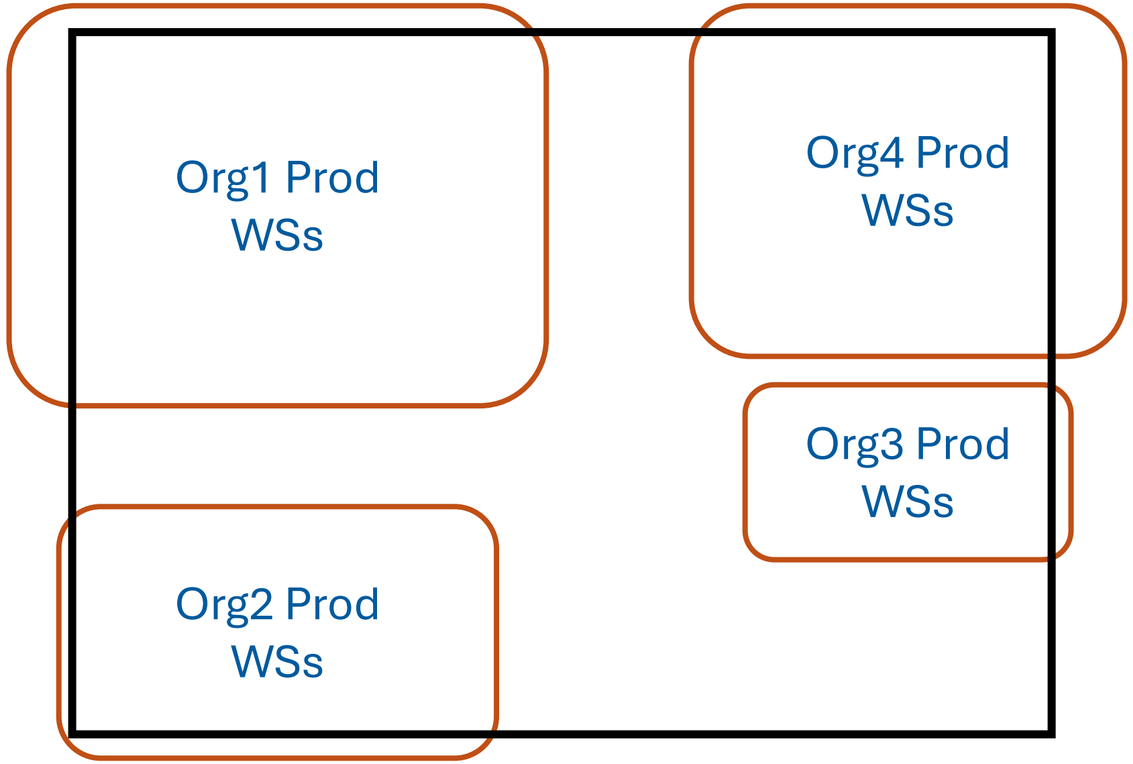

WSs = Workspaces

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Options

- Create multiple smaller F SKUs based on organization, type of work, etc.

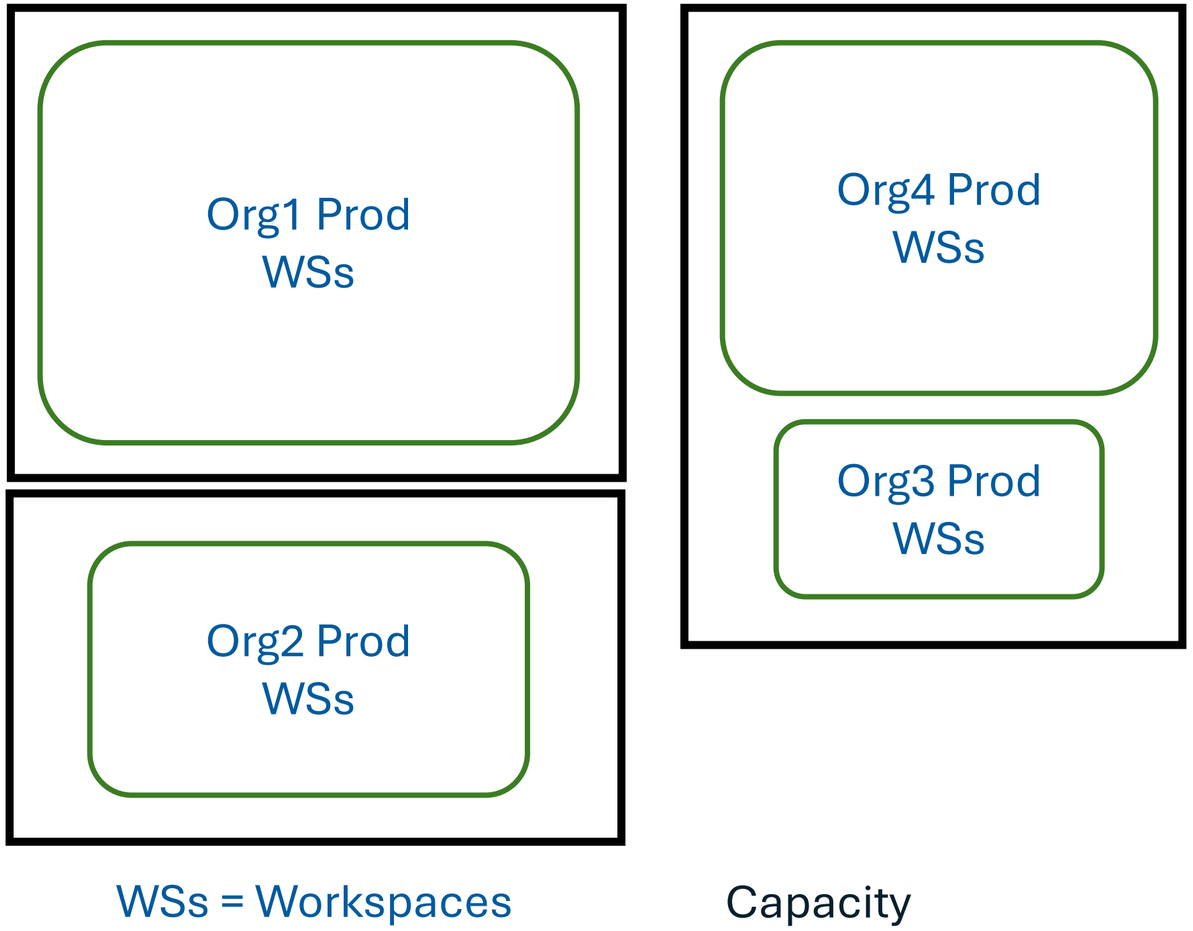
Pros

- Easy
- Provides some isolation from noisy neighbors (items with unintentionally high CU burn)
- Flexibility in capacity settings/governance

Cons

- Cost
- High CU items have increased chance of throttling
- Workload thresholds should be observed

Scale Out



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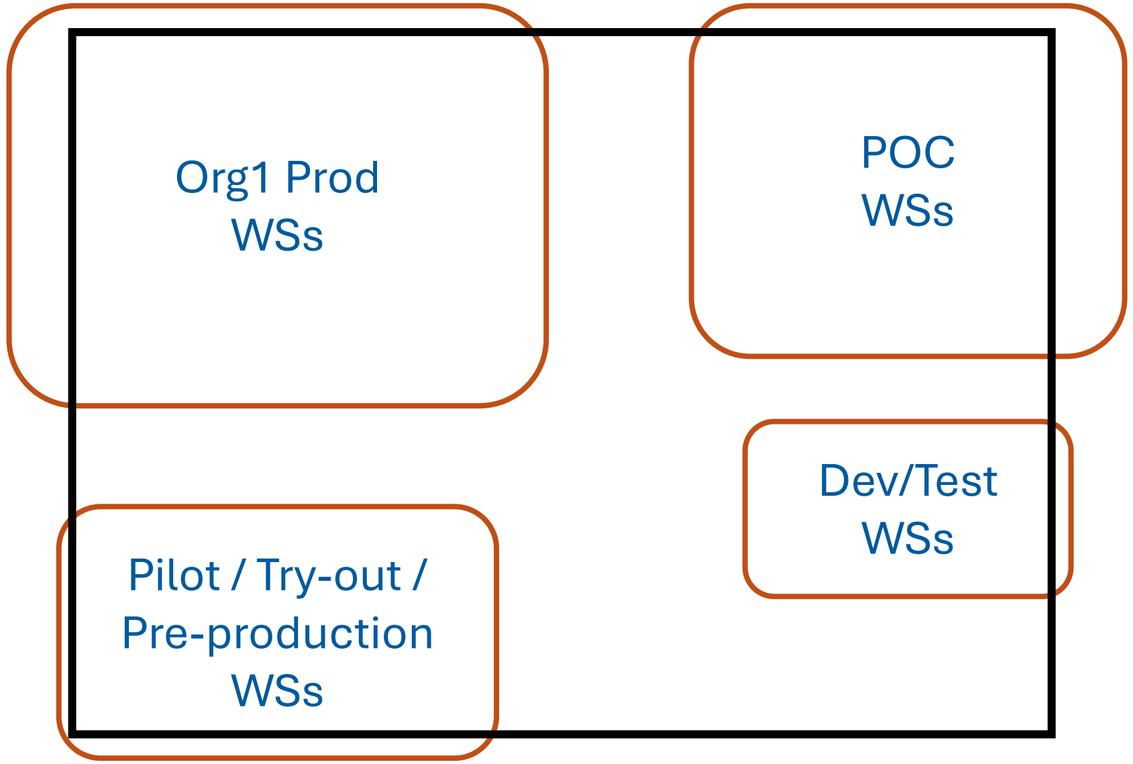


WSs = Workspaces

Capacity

Approach

- Provide isolated capacity for key items built by experienced developers

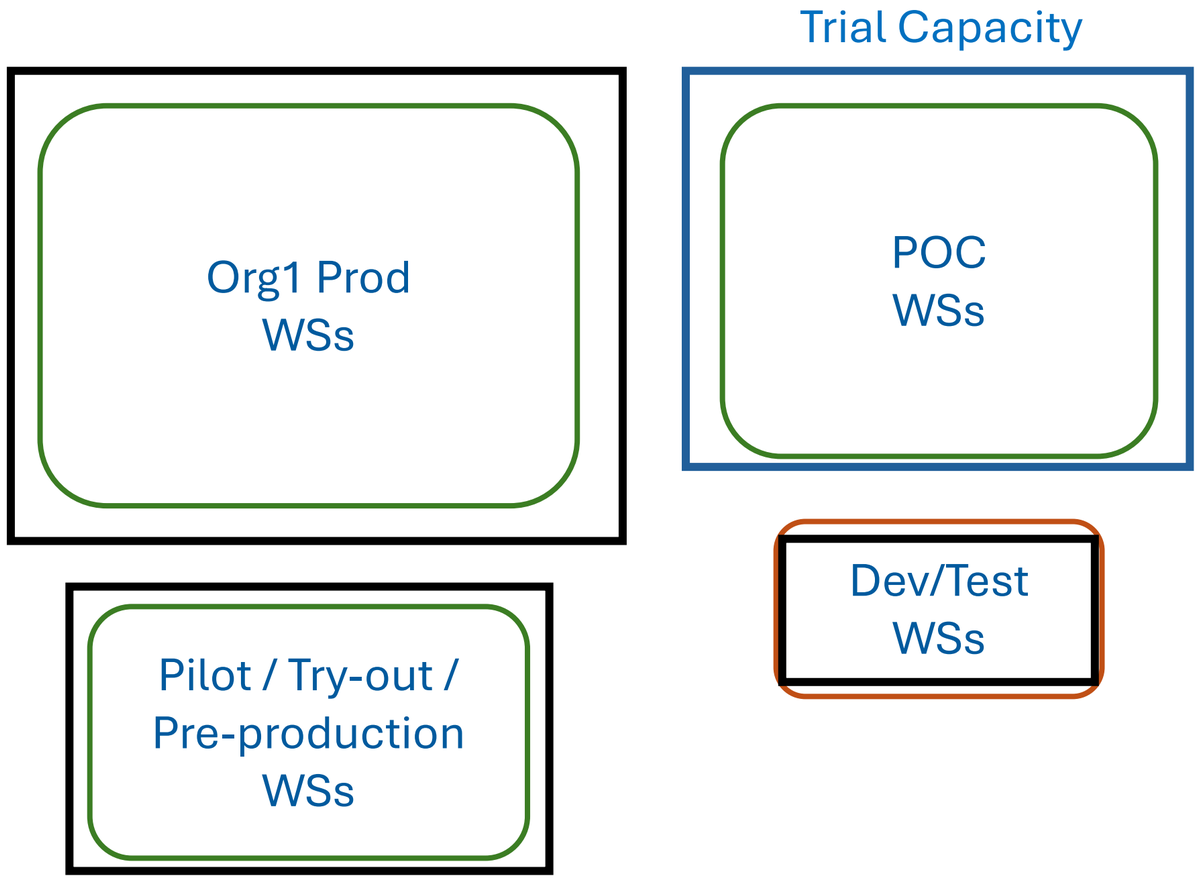
Pros

- Easy
- Provides isolation from items built by inexperienced developers and/or rapid unplanned usage growth
- Flexibility in capacity settings/governance

Cons

- Cost
- May lead to frustration of lower priority content developers/consumers

Isolate



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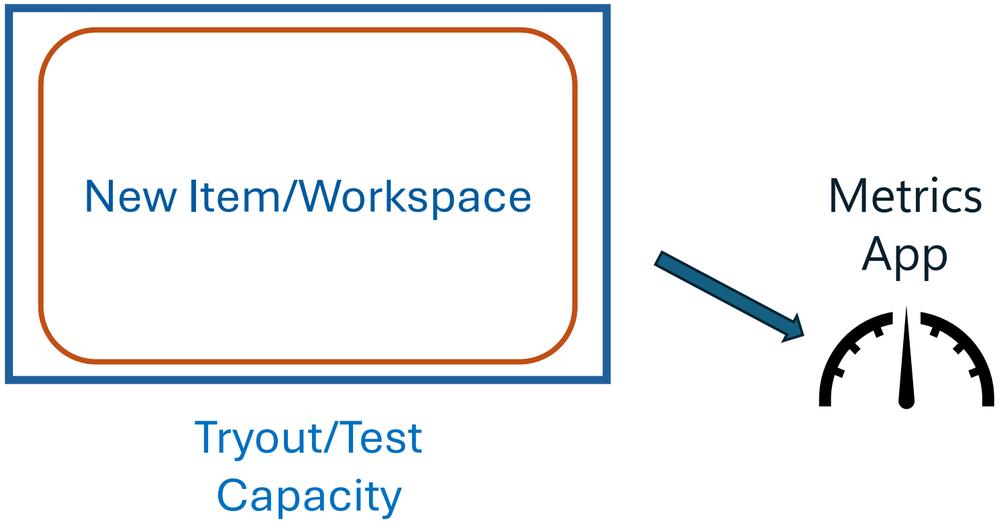
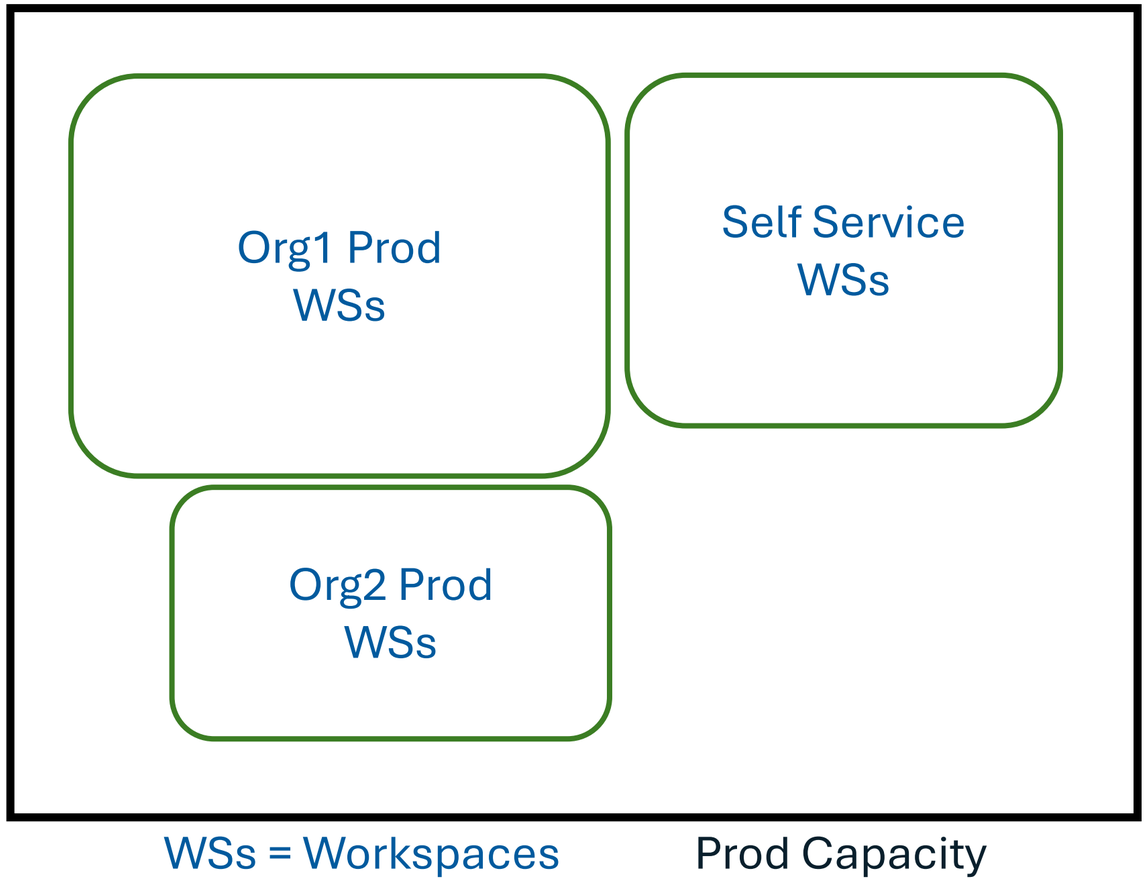
Isolation Strategy – Try-Out Capacity

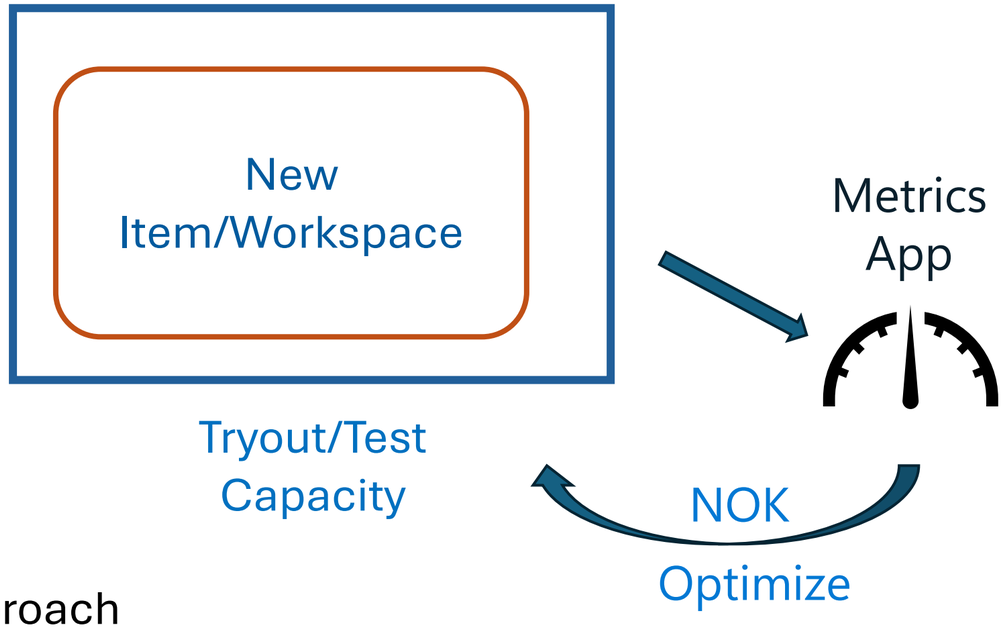


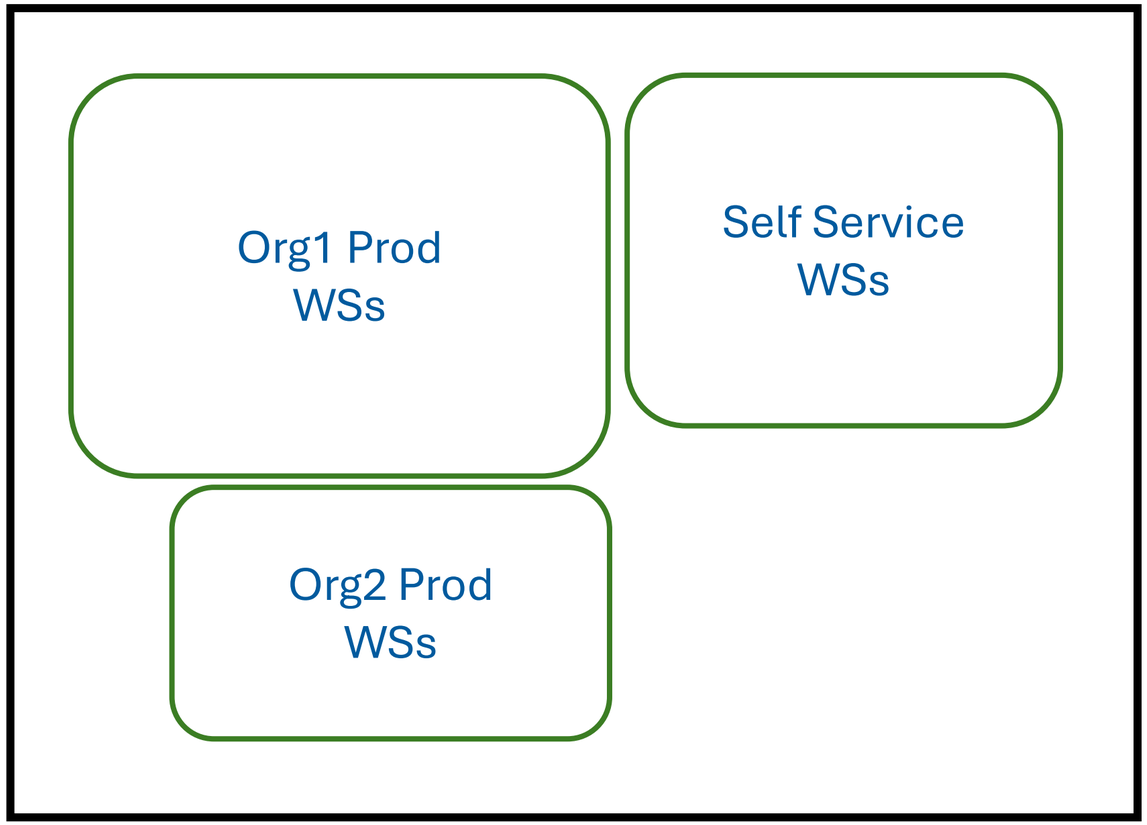


Approach

- Create a small F SKU capacity to **try out** new workspaces/items
- Assess CU consumption using metrics app
- If acceptable, move to production capacity
- If not, optimize
- Pause tryout capacity when not in use, if possible

Isolation Strategy – Try-Out Capacity





WSs = Workspaces

Prod Capacity

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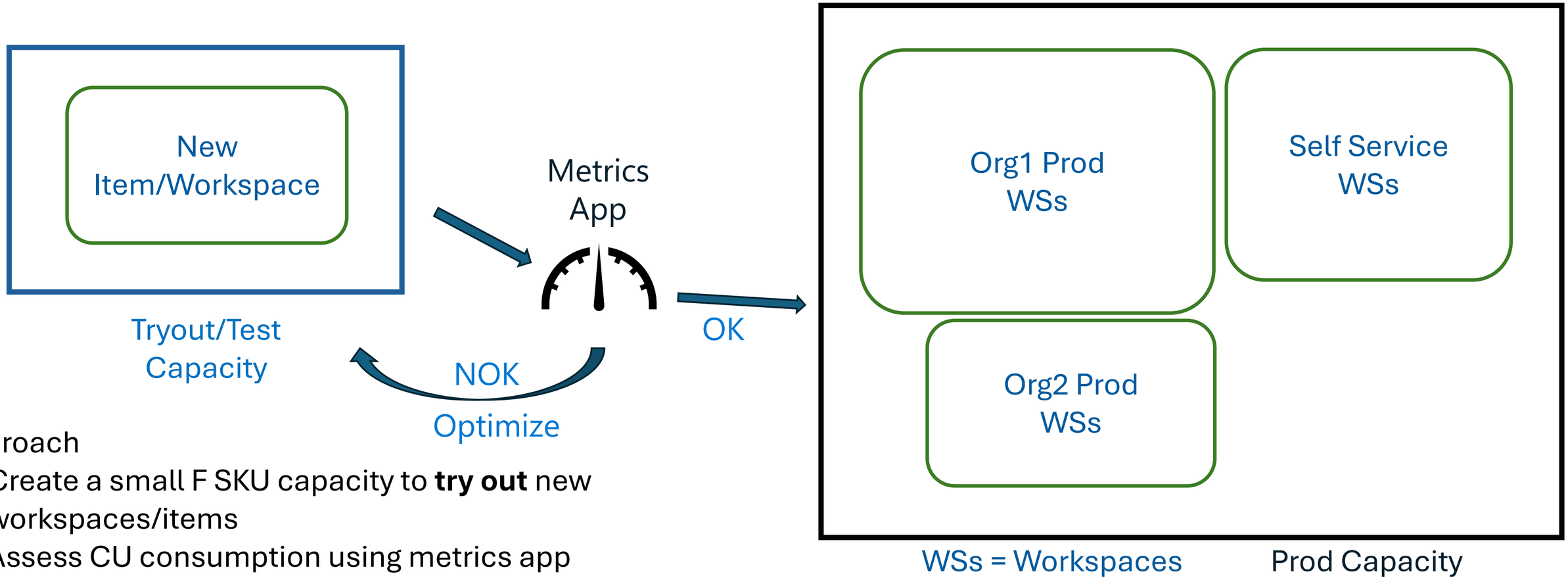
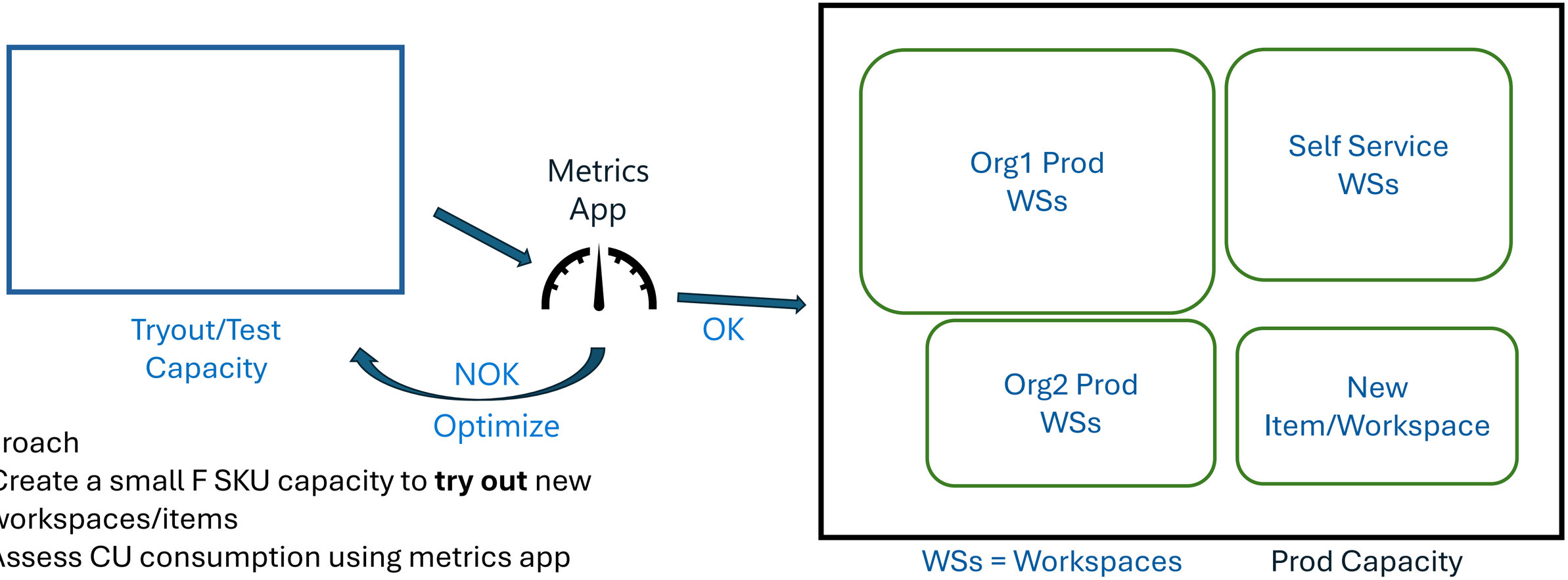
Isolation Strategy – Try-Out Capacity



Approach

- Create a small F SKU capacity to **try out** new workspaces/items
- Assess CU consumption using metrics app
- If acceptable, move to production capacity
- If not, optimize
- Pause tryout capacity when not in use, if possible

Isolation Strategy – Try-Out Capacity



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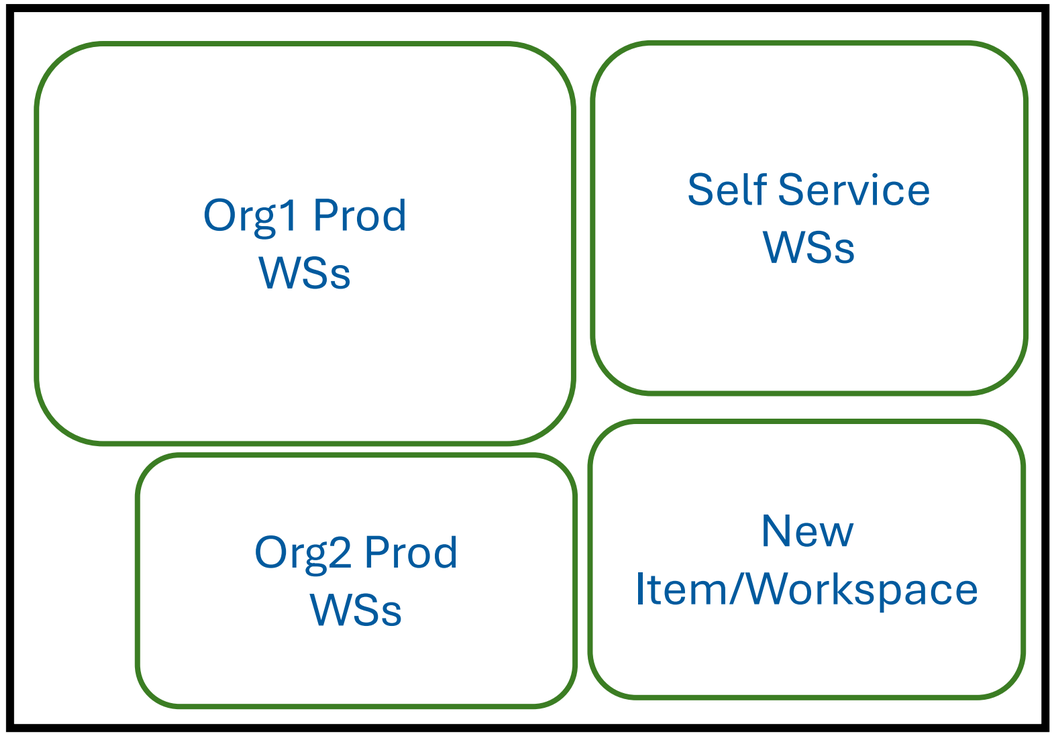
Isolation Strategy – Timeout Capacity

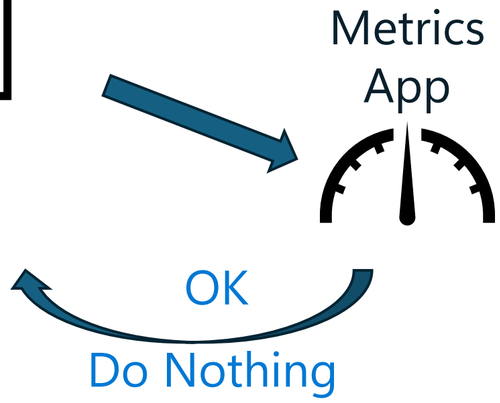


WSs = Workspaces

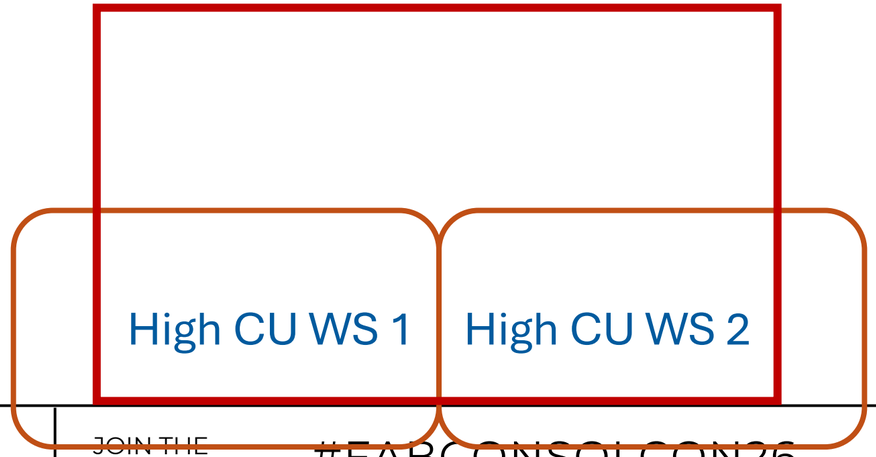
Prod Capacity

Approach

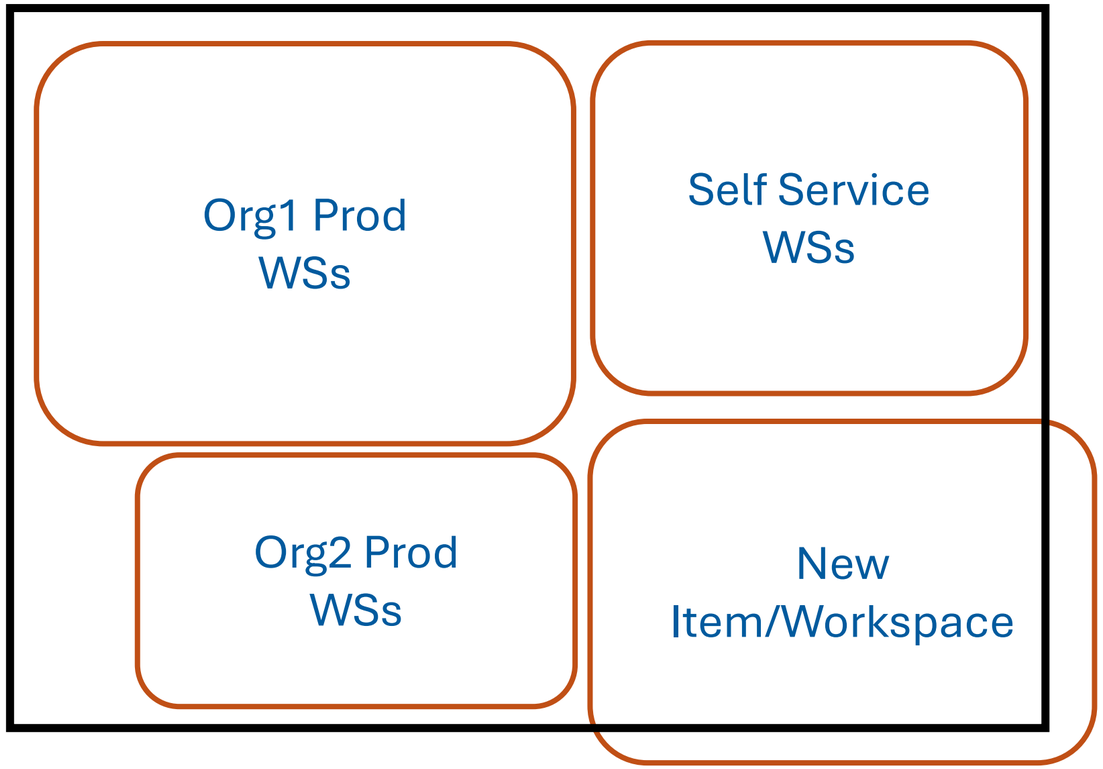
- Create a small F SKU capacity
- Assess CU consumption using metrics app
- If CU for new items/workspaces affects existing workloads (throttling), move new workspace to timeout capacity (Admin Portal/Capacity Settings)
- High CU items/WSs (noisy neighbors) share smaller capacity (or you can pause it post move)
- Note workload thresholds before moving



Timeout Capacity



Isolation Strategy – Timeout Capacity



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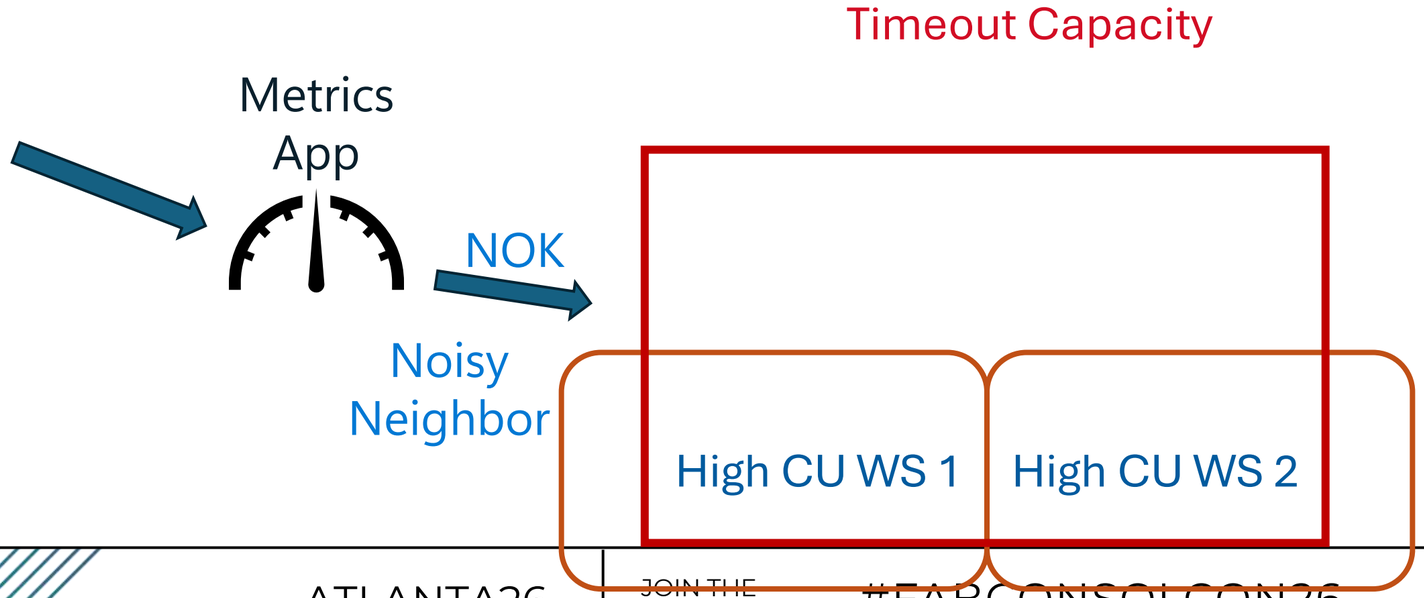
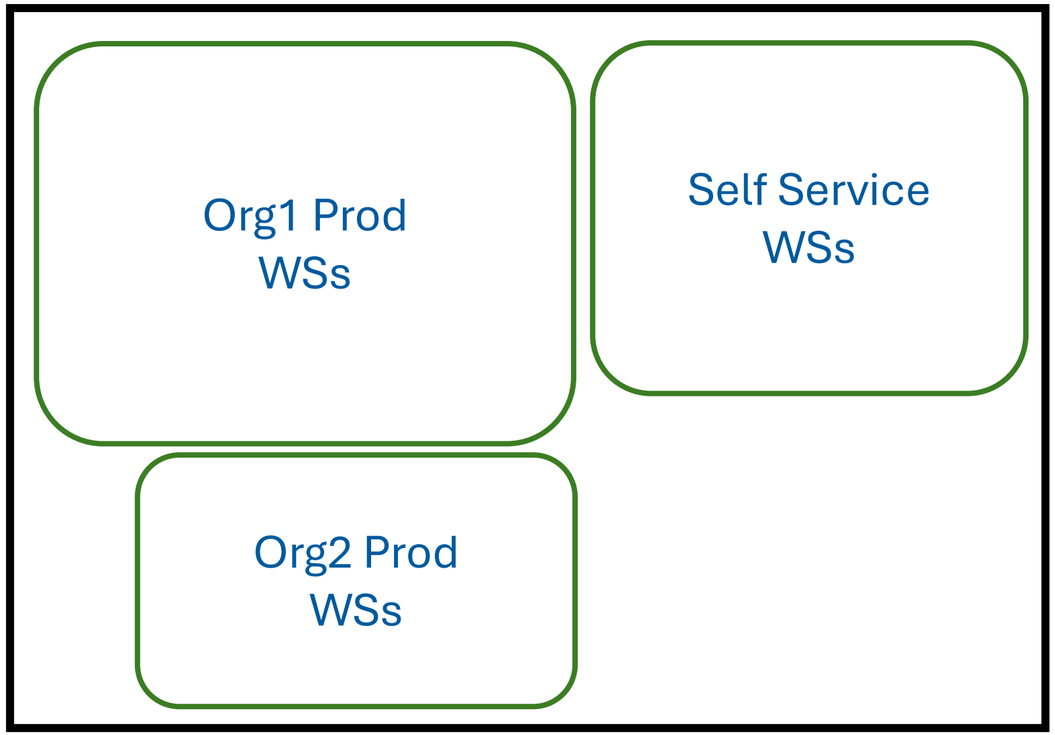


Isolation Strategy – Timeout Capacity



WSs = Workspaces

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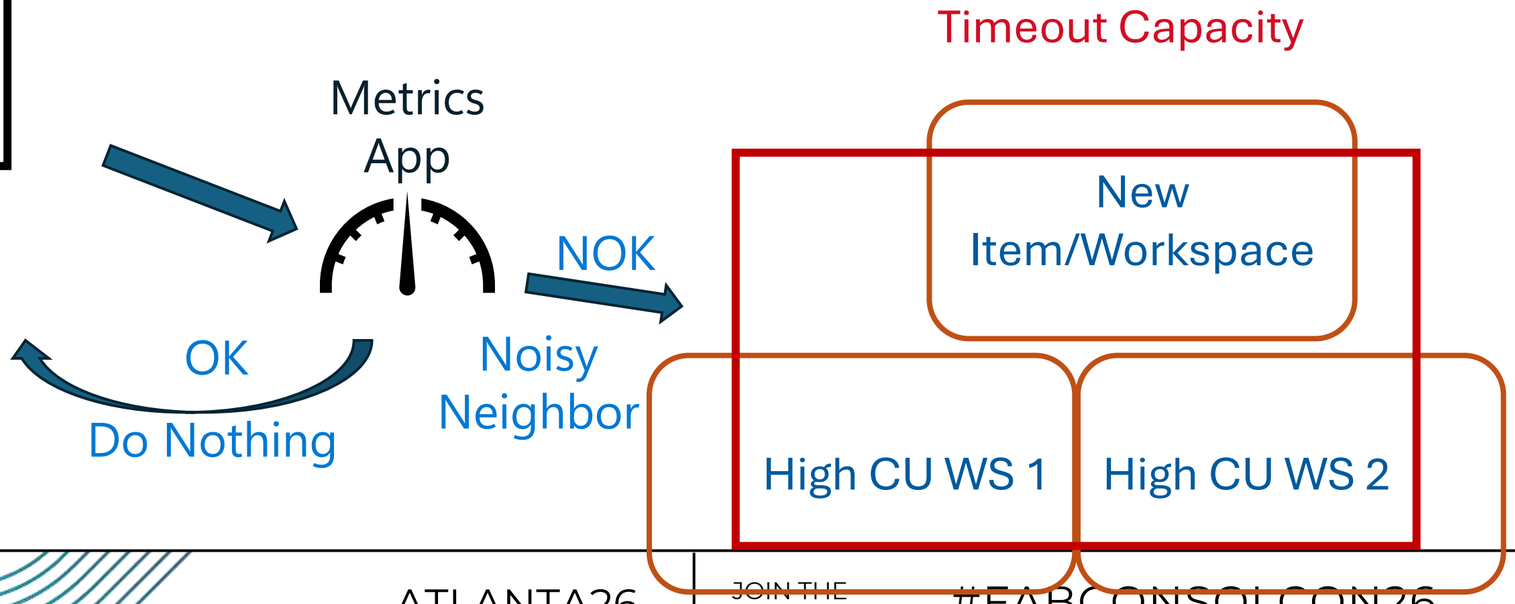
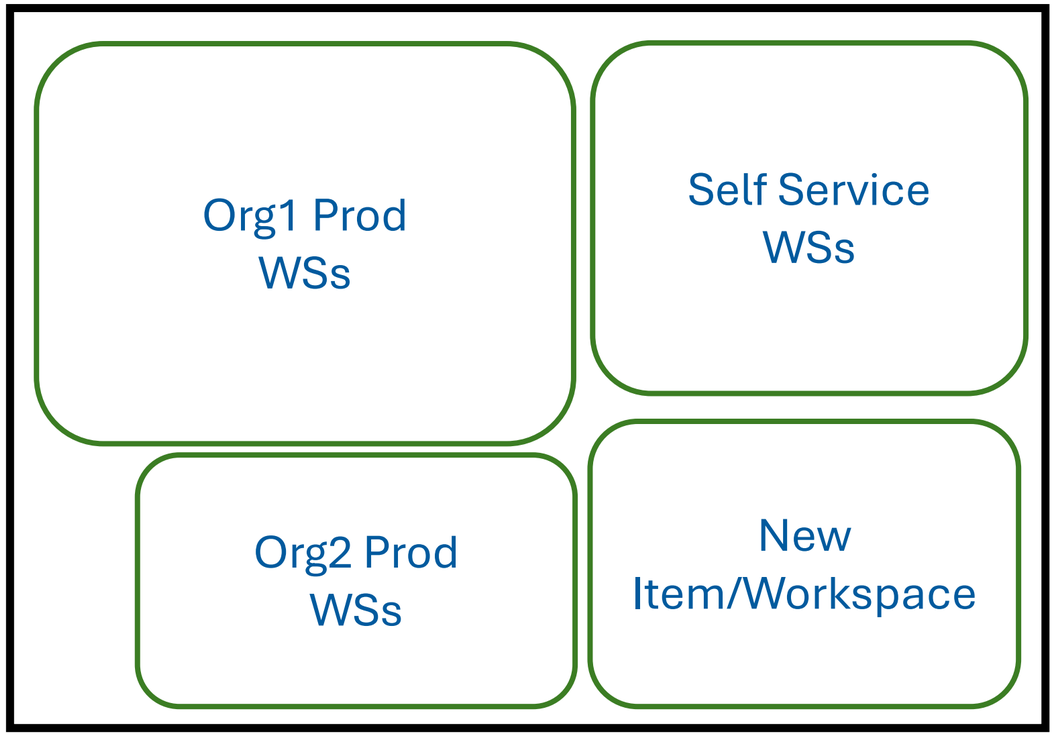


Isolation Strategy – Rescue Capacity

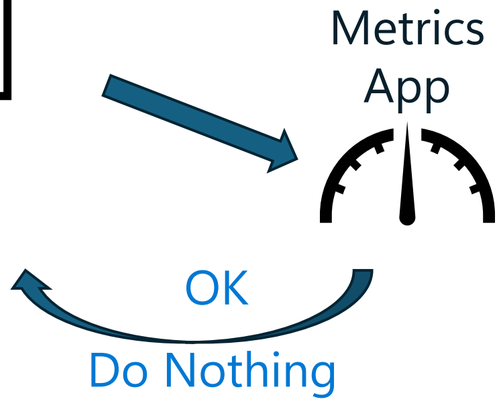


WSs = Workspaces

Prod Capacity

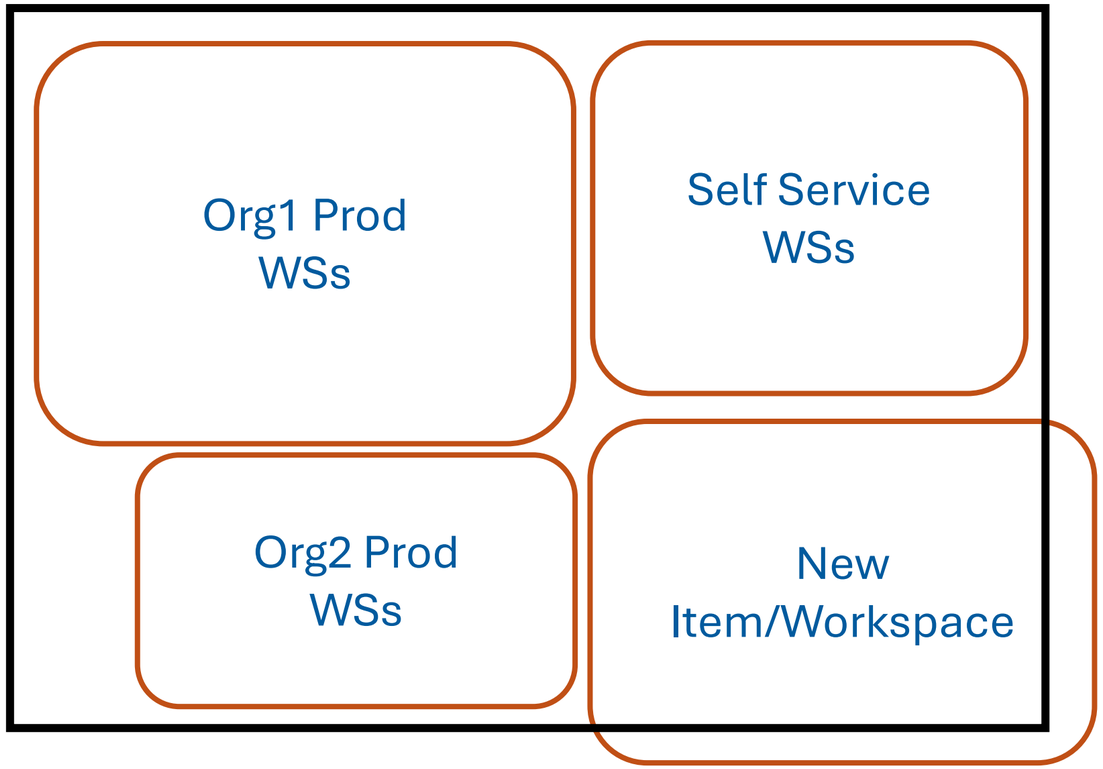
Approach

- Create an F SKU capacity, keep it paused
- Assess CU consumption using metrics app
- If CU for new items/workspaces affects priority workloads (throttling), resume the new capacity
- Move priority workspace to new capacity (Admin Portal/Capacity Settings)
- Address issues with new content, then bring it back to original capacity, and pause the new one
- Note workload thresholds before moving



Rescue Capacity

Isolation Strategy – Rescue Capacity



WSs = Workspaces

Prod Capacity

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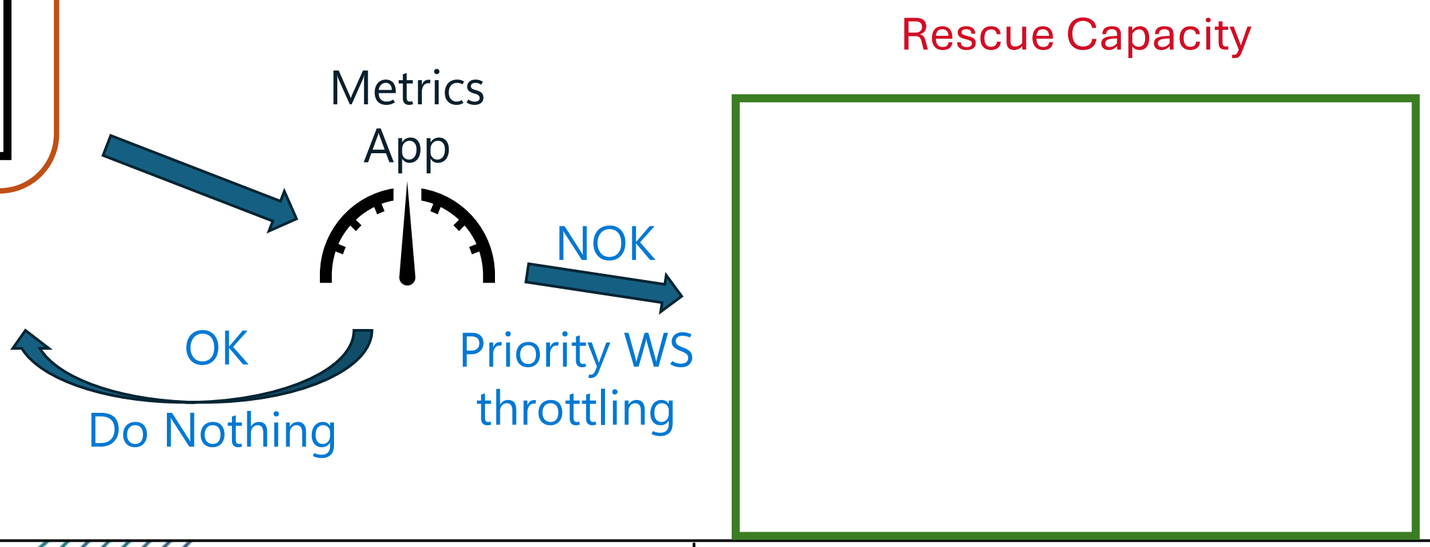

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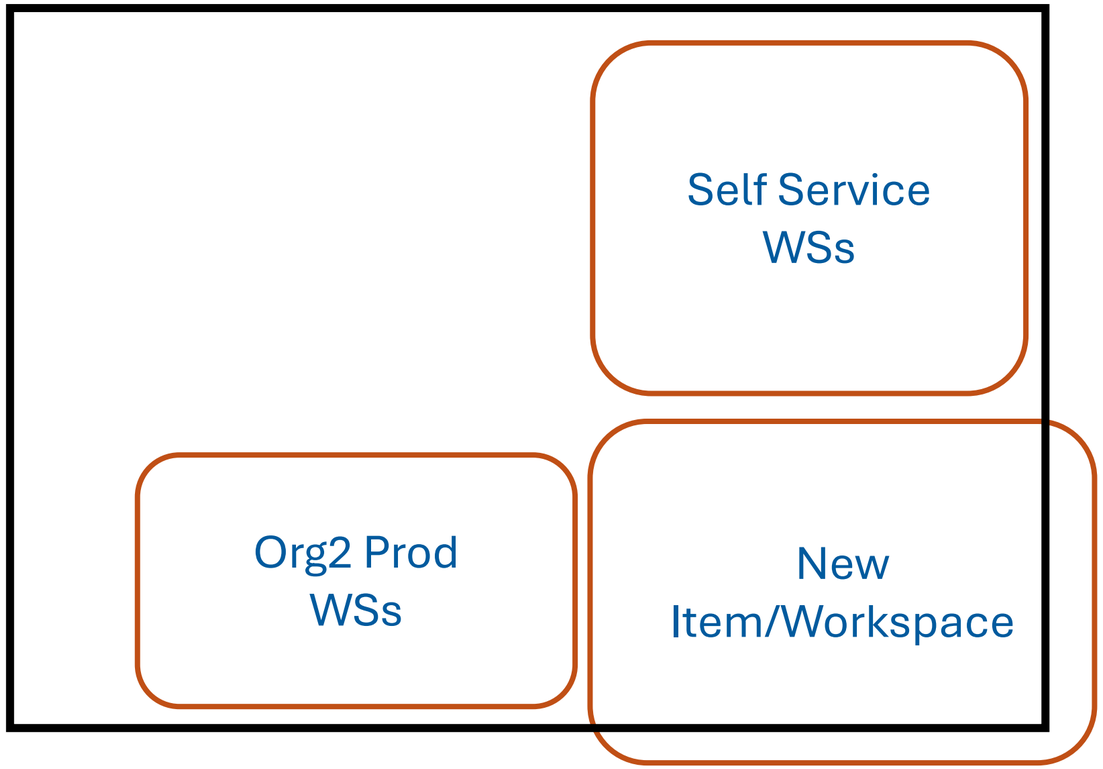

WSs = Workspaces

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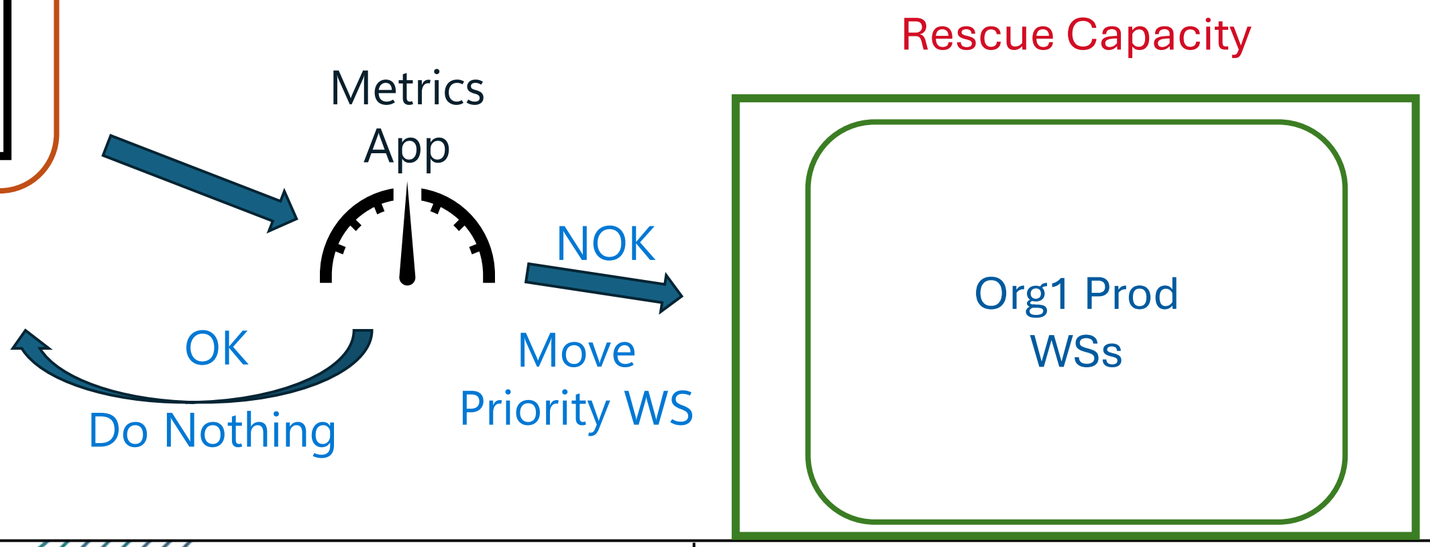


Recommended Configurations – Typical (your mileage varies)

Dev / Test

- PayGo Or smaller Shared capacity
- Surge Protection at Capacity (v1)
- Surge Protection at Workspace (v2)
- Timeout Capacities
- Try out Capacities

UAT / Shared Prod

- Reserved Capacity
- Surge Protection at Capacity (v1)
- Surge Protection at Workspace (v2)
- Hi-Pri workspace
  - Mission Critical
  - Overage Protection
  - RTH for troubleshooting

Central Prod – Serving Layer

- Reserved Instance with headroom
- Capacity level surge protection (optional)
- Mission Critical
- RTH and other logging Enabled
- Overage Protection

Central Prod Processing / Store Layer

- Reserved Instance
- Mission Critical
- Overage protection
- RTH and other logging enabled
- Autoscale for Spark

Sound off.
The mic is all yours.
Influence the product roadmap.

Join the Fabric User Panel

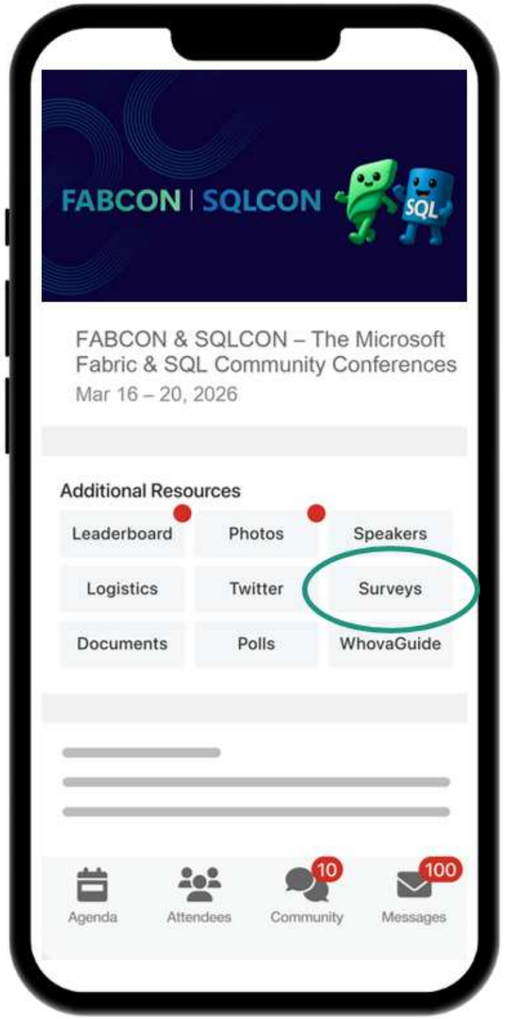
Share your feedback directly with our Fabric product group and researchers.

https://aka.ms/JoinFabricUserPanel

Join the SQL User Panel

Influence our SQL roadmap and ensure it meets your real-life needs

https://aka.ms/JoinSQLUserPanel

How was the session?

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